This demo contains:

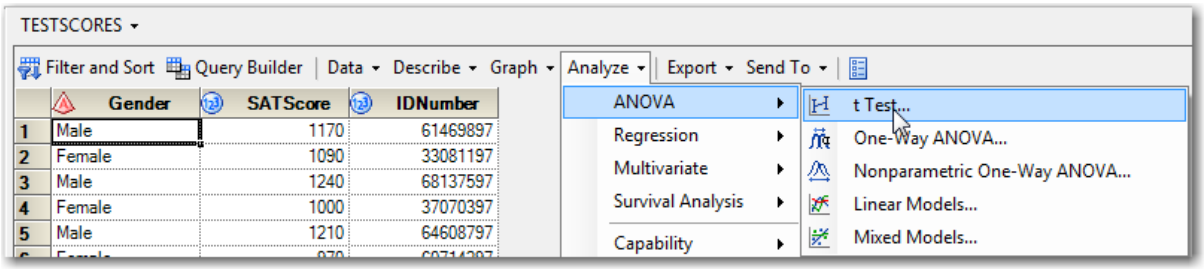
- **Two-Sample *t*-Test**
- **Descriptive Statistics across Groups**
- **The Linear Models Task**
- **ANOVA with Blocking**
- **Post Hoc Pairwise Comparisons**
- **Two-Way ANOVA with Interactions**

Two-Sample *t*-Test

Perform a two-sample t-test comparing girls to boys on SAT Math + Reading mean score, using the t Test task.

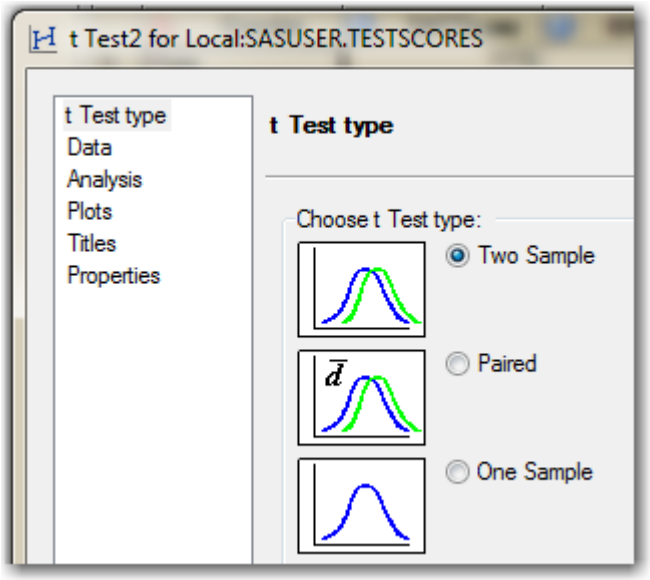
First it is advisable to verify the assumptions of *t*-tests. There is an assumption of normality of the distribution of each group. This assumption can be verified with a quick check of the Summary Panel and the Q-Q Plot.

1. Create a new process flow and rename it **Analysis of Variance**.
2. Open the **TESTSCORES** data set from the SASUSER library on the Local server.
3. Select **Analyze** ⇨ **ANOVA** ⇨ **t Test**.

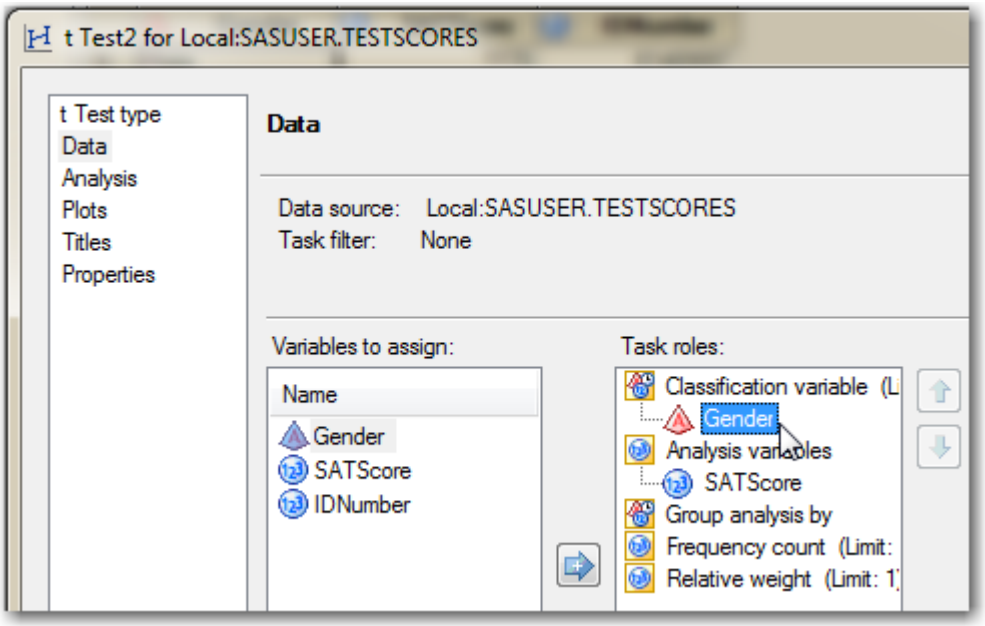


Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

4. Leave **Two Sample** selected.

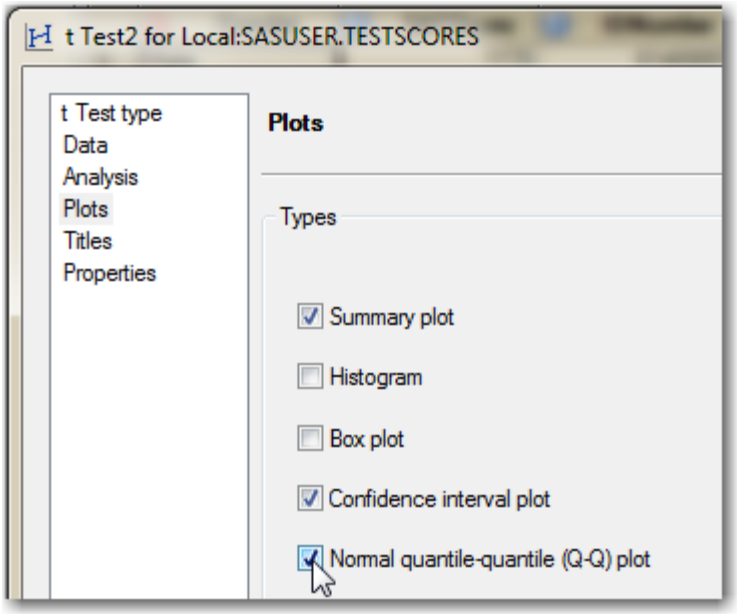


5. Under **Data**, select **SATScore** as the analysis variable task role and **Gender** as the classification variable.



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

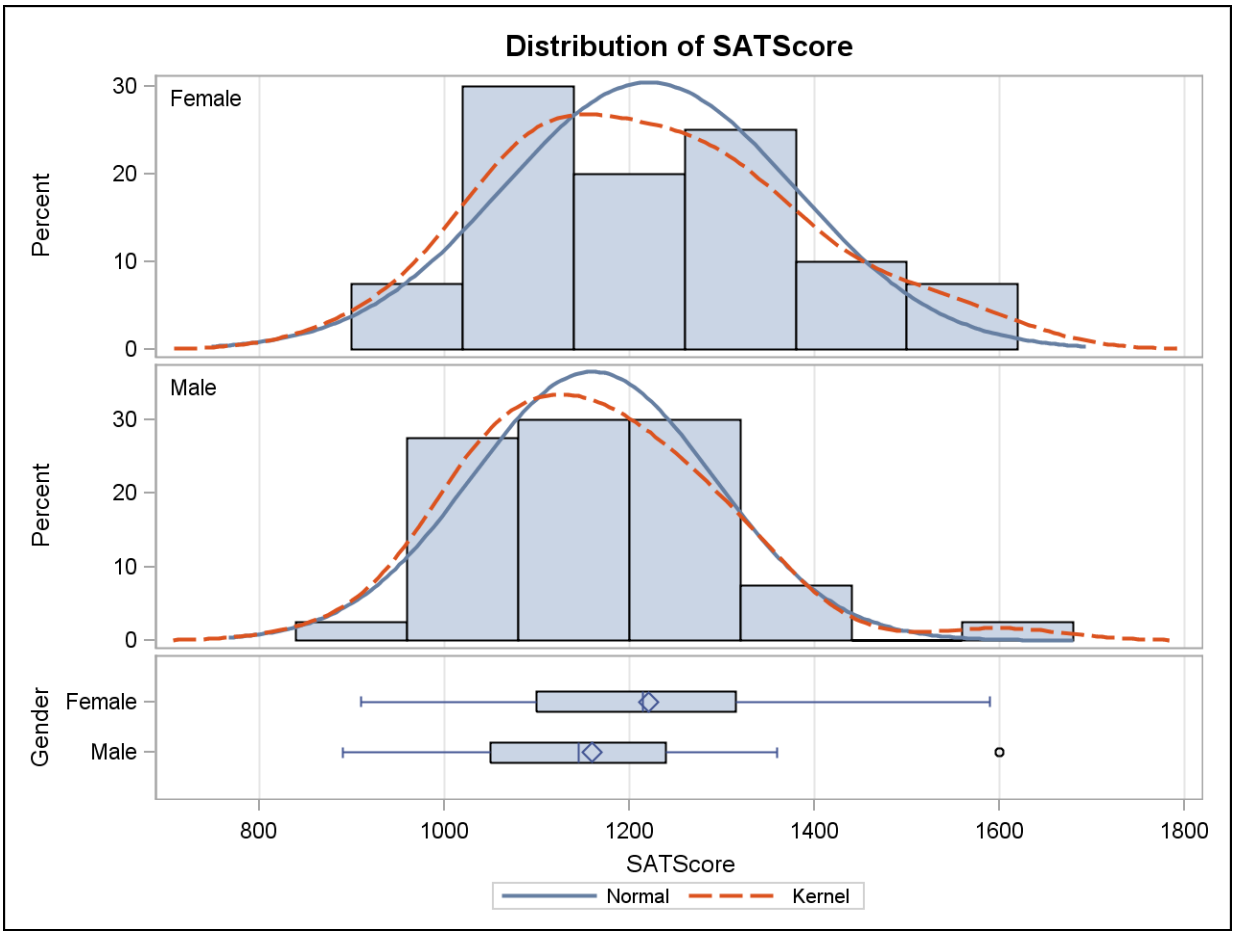
6. Under **Plots**, check **Summary plot**, **Confidence interval plot**, and **Normal quantile-quantile (Q-Q) plot**.

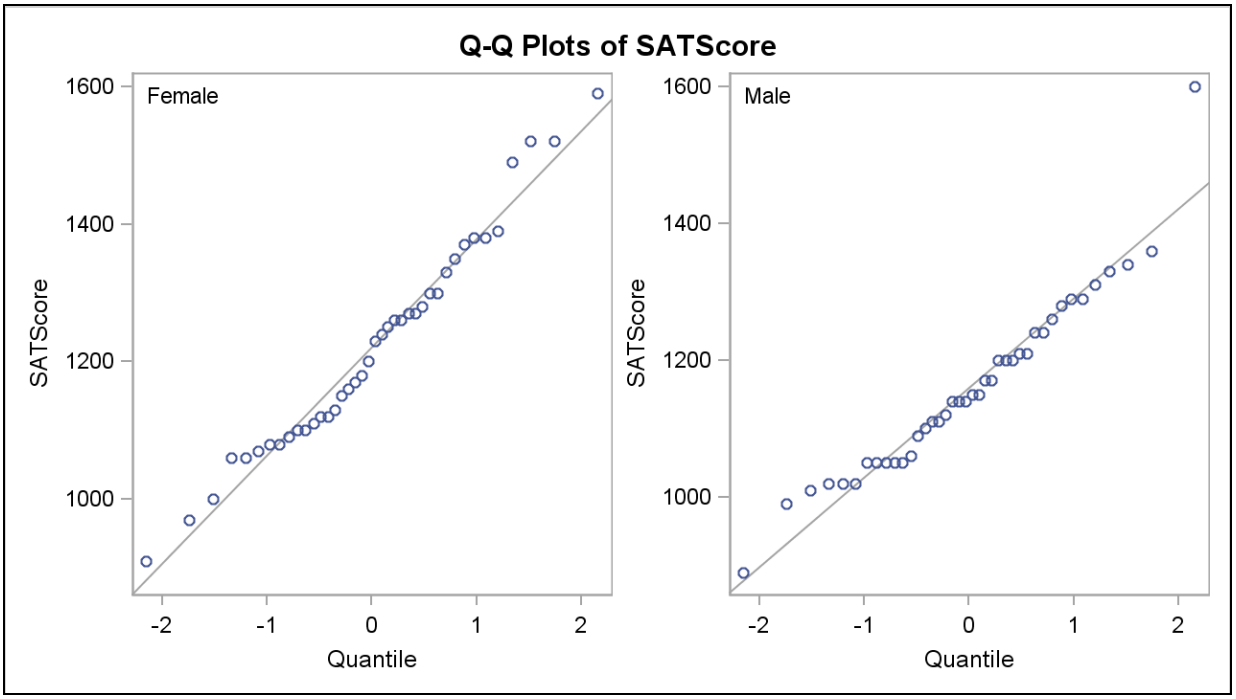


7. Change the titles under **Properties**, if desired, and then click **Run**.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

First look at plots of the distribution of **SATScore** for each of the genders.





Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

The Q-Q Plot is similar to the Normal Probability Plot produces in the Distribution Analysis task. The X-axis for this plot is scaled as quantiles, rather than probabilities. For each group it seems that the data approximate a normal distribution. There seems to be one potential outlier – a male scoring a perfect 1600 on the SAT, when no other male scored greater than 1400.

If assumptions are not met, you can do an equivalent nonparametric test, which does not make distributional assumptions. It is described in an appendix.

The statistical tables for the t Test task are displayed below.

❶

Gender	N	Mean	Std Dev	Std Err	Minimum	Maximum
Female	40	1221.0	157.4	24.8864	910.0	1590.0
Male	40	1160.3	130.9	20.7008	890.0	1600.0
Diff (1-2)		60.7500	144.8	32.3706		

Gender	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Female		1221.0	1170.7	1271.3	157.4	128.9	202.1
Male		1160.3	1118.4	1202.1	130.9	107.2	168.1
❸ **Diff (1-2)**	**Pooled**	60.7500	-3.6950	125.2	144.8	125.2	171.7
Diff (1-2)	**Satterthwaite**	60.7500	-3.7286	125.2			

Method	Variances	DF	t Value	Pr > |t|
❸ **Pooled**	Equal	78	1.88	0.0643
Satterthwaite	Unequal	75.497	1.88	0.0644

Equality of Variances				
Method	**Num DF**	**Den DF**	**F Value**	**Pr > F**
❷ **Folded F**	39	39	1.45	0.2545

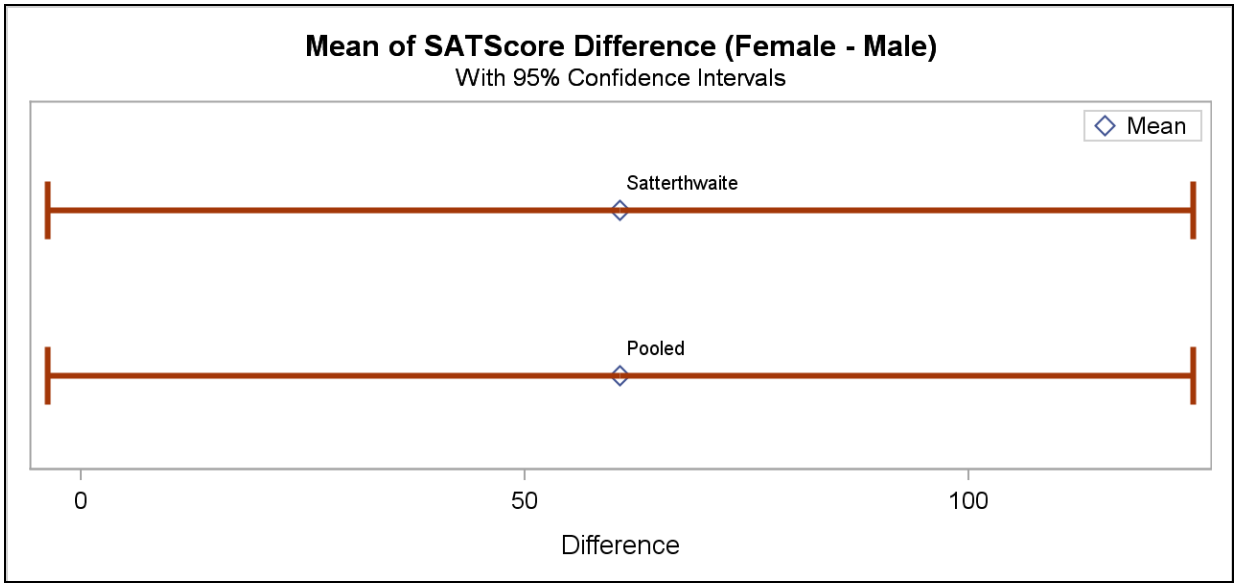
❶ In the Statistics table, examine the descriptive statistics for each group and their differences.

❷ Look at the Equality of Variances table that appears at the bottom of the output. The *F*-test for equal variances has a *p*-value of 0.2545. Because this value is greater than the alpha level of 0.05, do not reject the null hypothesis of equal variances (This is equivalent to saying that there is insufficient evidence to indicate that the variances are not equal.)

❸ Based on the *F*-test for equal variances, you then look in the T-Tests table at the *t*-test for the hypothesis of equal means. Using the equal variance (Pooled) *t*-test, you do not reject the null hypothesis that the group means are equal. The mean difference between boys and girls is 60.75. However, because the *p*-value is greater than 0.05 (Pr > |t| = 0.0643), you conclude that there is no significant difference in the average SAT score between boys and girls.

The confidence interval for the mean difference (-3.6950, 125.2) includes 0. This implies that you cannot say with 95% confidence that the difference between boys and girls is ***not*** zero. Therefore, it also implies that the *p*-value is greater than 0.05.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.



Confidence intervals are shown in the Difference interval plot. Because the variances here are so similar between males and females, the Pooled and Satterthwaite intervals (and *p*-values) are very similar. The lower bound of the Pooled interval extends past zero.

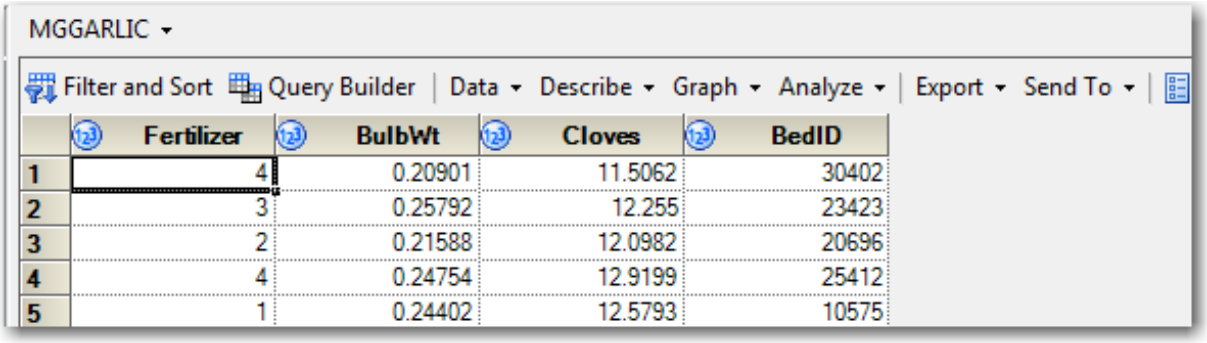
A good argument can be made that the point estimate for the difference between males and females is big from a practical standpoint. If the sample were a bit larger, that same difference might be significant because the pooled standard error would be smaller. However, such a sample needs to be drawn to confirm this hypothesis.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Descriptive Statistics across Groups

Obtain summary statistics and box plots for the **MGGARLIC** data set.

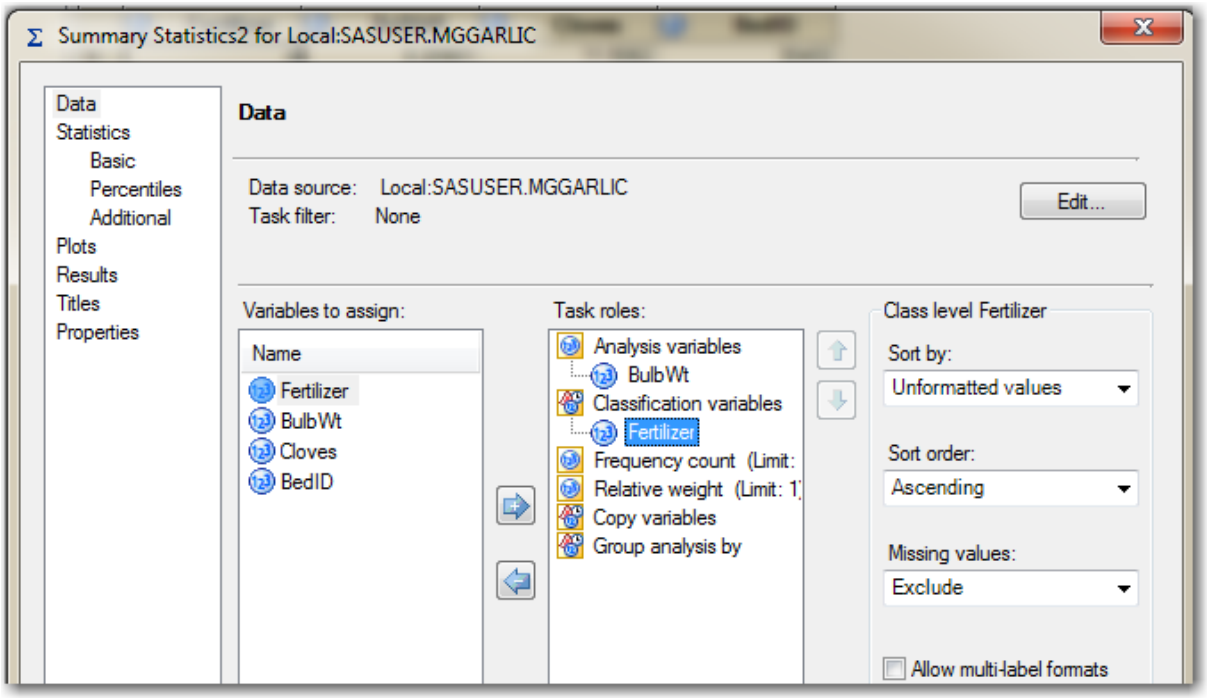
1. Open the **MGGARLIC** data set in the Analysis of Variance process flow.



	Fertilizer	BulbWt	Cloves	BedID
1	4	0.20901	11.5062	30402
2	3	0.25792	12.255	23423
3	2	0.21588	12.0982	20696
4	4	0.24754	12.9199	25412
5	1	0.24402	12.5793	10575

2. Select **Describe** ⇨ **Summary Statistics**.

3. Under **Data**, select **BulbWeight** as the analysis variable and **Fertilizer** as the classification variable.



4. Under **Plots**, check **Box and whisker**.

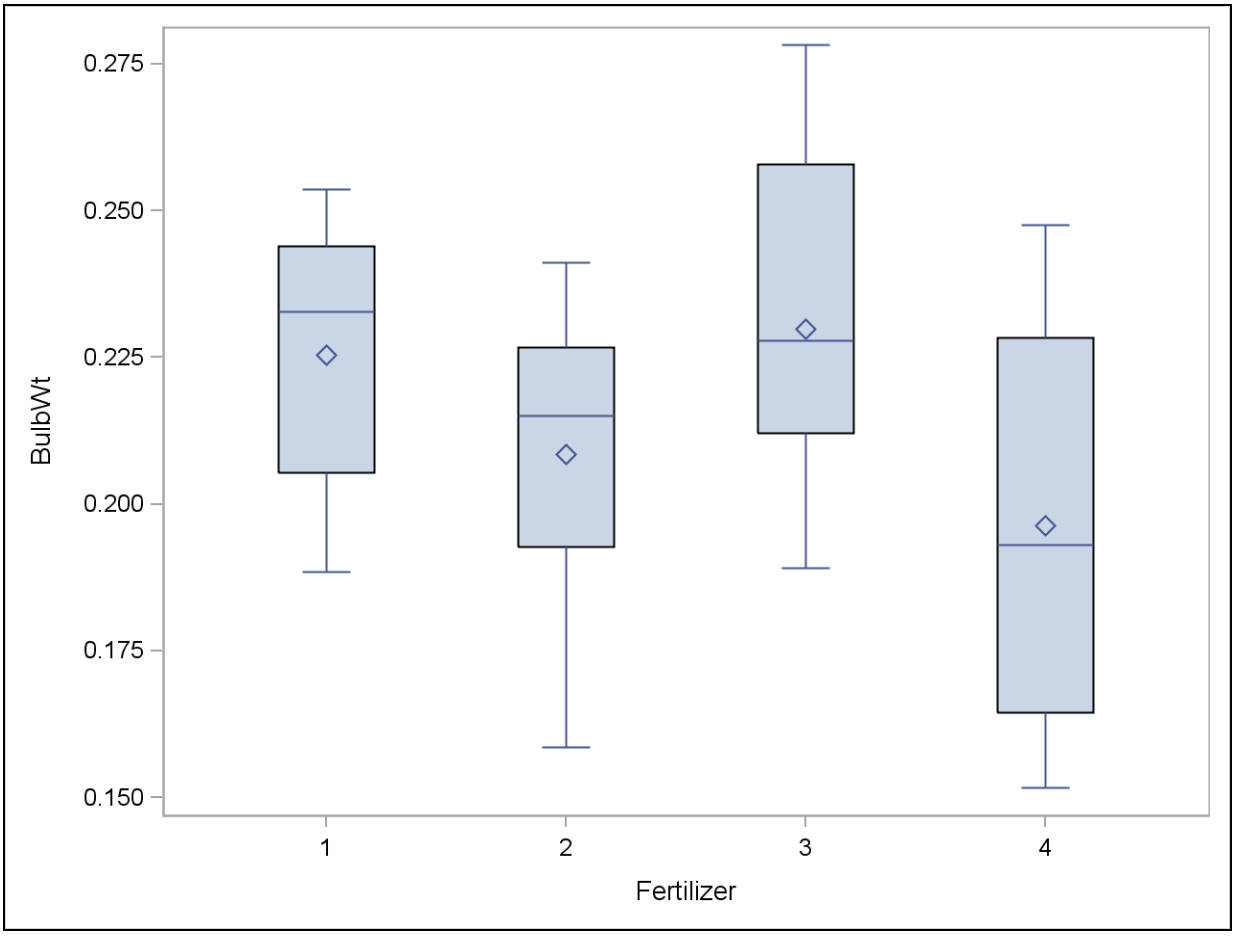
5. Under **Basic**, change the **Maximum decimal places** to **3**.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

6. Change the titles as desired and then click **Run**.

Analysis Variable : BulbWt						
Fertilizer	N Obs	Mean	Std Dev	Minimum	Maximum	N
1	9	0.225	0.025	0.188	0.254	9
2	8	0.209	0.026	0.159	0.241	8
3	11	0.230	0.026	0.189	0.278	11
4	4	0.196	0.041	0.152	0.248	4

The design is ***not balanced***. In other words, the groups are not equally sized.



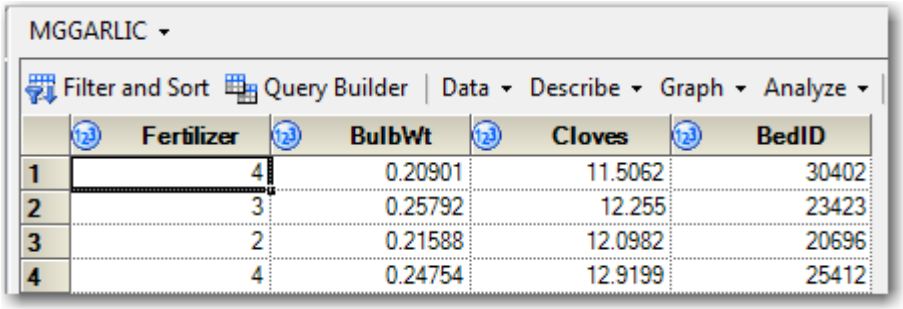
The box plots do not provide much additional information. There are simply too few observations per group for them to be very informative.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

The Linear Models Task

Perform ANOVA to test whether the mean bulb weight of garlic is different across different fertilizers.

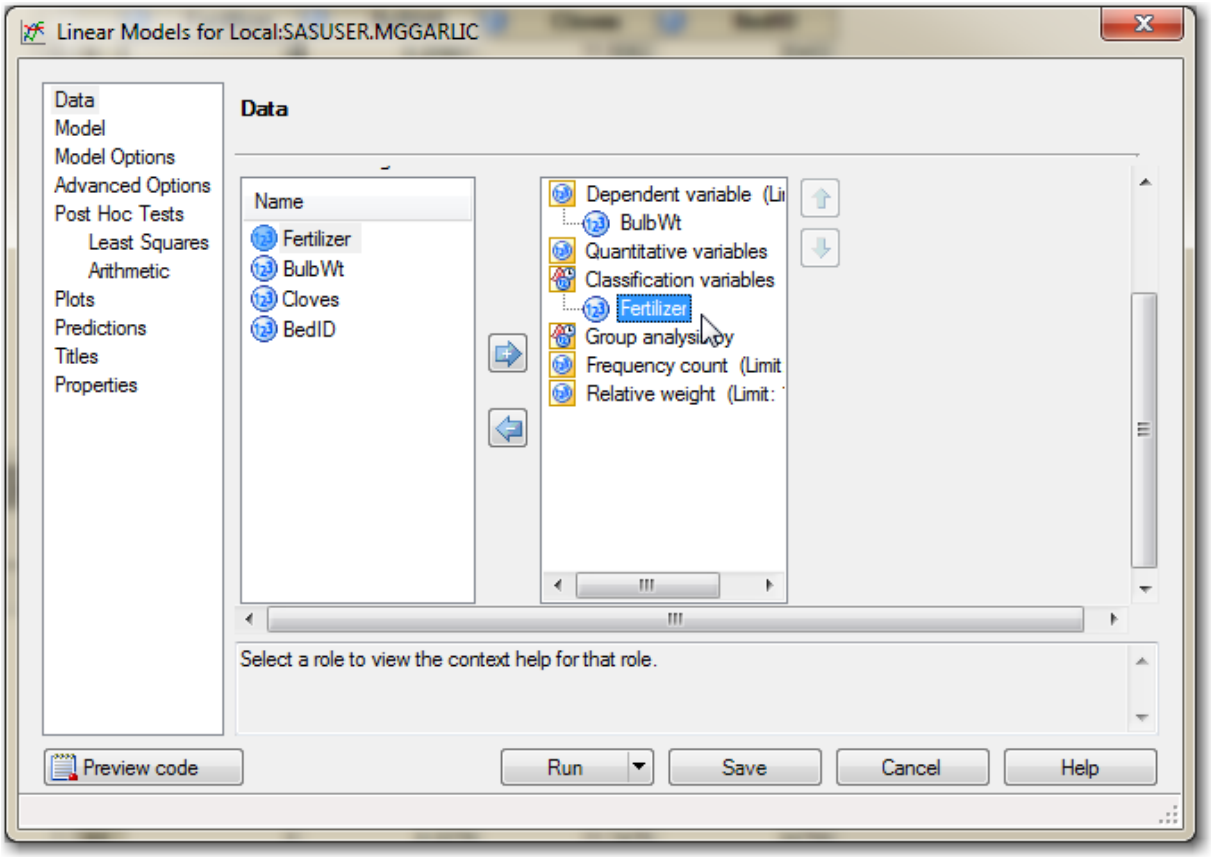
1. Click the **Input Data** tab in the Analysis of Variance task flow to expose the **MGGARLIC** data set.



	Fertilizer	BulbWt	Cloves	BedID
1	4	0.20901	11.5062	30402
2	3	0.25792	12.255	23423
3	2	0.21588	12.0982	20696
4	4	0.24754	12.9199	25412

2. Select **Tasks** ⇨ **ANOVA** ⇨ **Linear Models**.

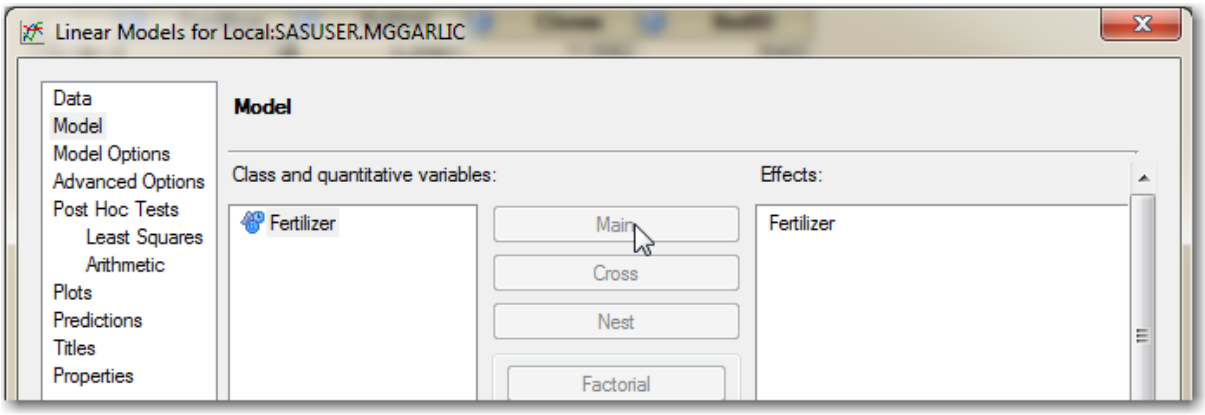
3. Under **Data**, assign **BulbWeight** and **Fertilizer** to the task roles of dependent variable and classification variable, respectively.



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

4. Under **Model**, select **Fertilizer** and then click **Main**.

   The selection of task roles populated the **Class and quantitative variables** pane.

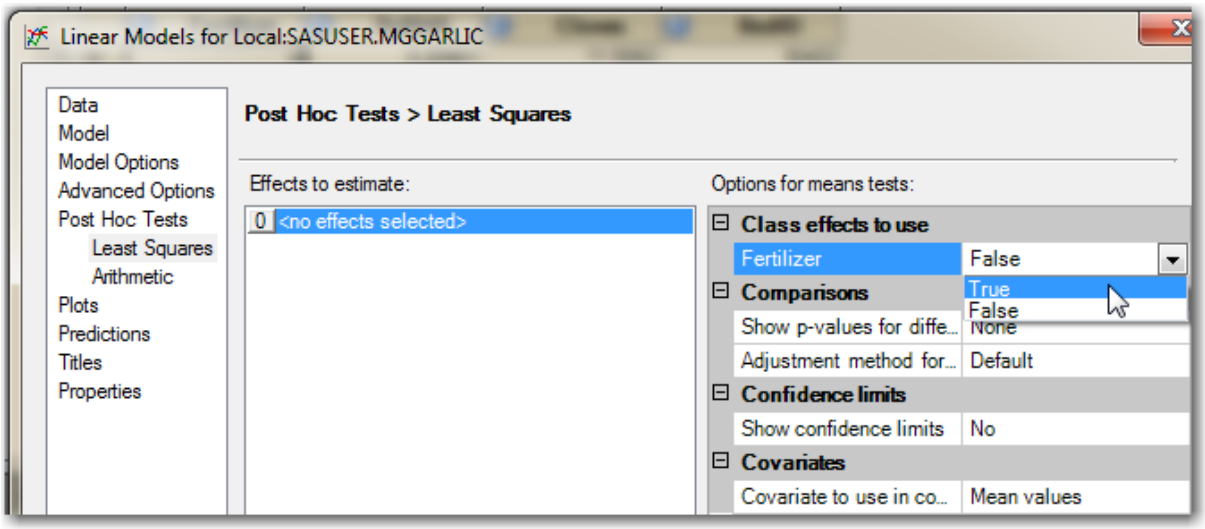


5. Under **Model Options**, uncheck **Type I** and **Show parameter estimates**.

6. Select **Least Squares** under **Post Hoc Tests** at the left.

7. Click **Add** at the bottom.

8. For **Class effects to use**, next to **Fertilizer**, click **False** twice to change that option to **True**.



9. Under **Comparisons**, for **Show p values for differences**, change the selection from **All** to **All pairwise differences**.

10. For **Adjustment method for comparison**, you can either keep the default or select **Tukey**, which is the default when all pairwise differences are selected.

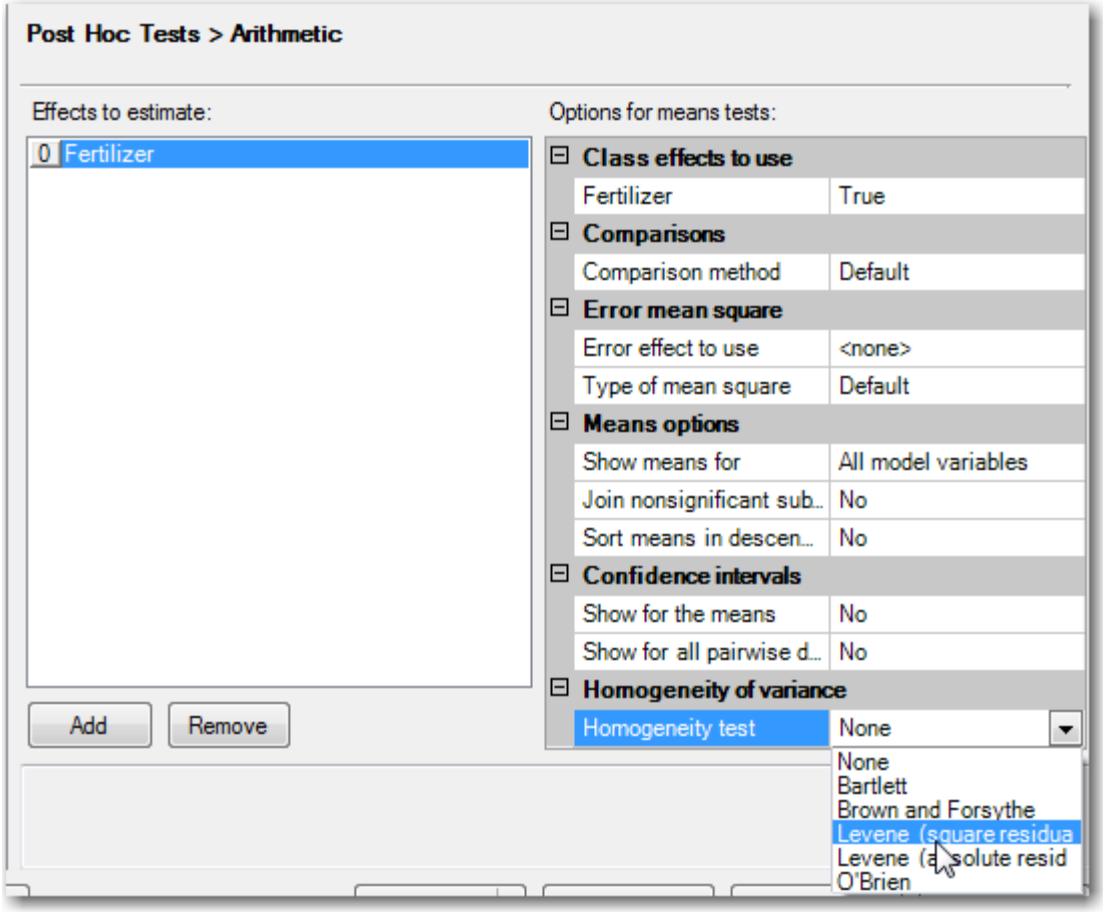
11. Select **Arithmetic** under **Post Hoc Tests** at the left.

12. Click **Add** at the bottom.

13. For **Class effects to use**, next to **Fertilizer**, click **False** twice to change that option to **True**.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

14. Under **Homogeneity of variance**, select **Levene (square residuals)**.



15. Click **Run**.

Turn your attention to the first two tables of the output. The first table specifies the number of levels and the values of the class variable.

Class Level Information		
Class	**Levels**	**Values**
Fertilizer	4	1 2 3 4

The second table shows both the number of observations read and the number of observations used. These values are the same because there are no missing values in for any variable in the model. If any row has ***missing data*** for a predictor or response variable, that row is ***dropped*** from the analysis.

Number of Observations Read	32
Number of Observations Used	32

The second part of the output contains all of the information that is needed to test the equality of the treatment means. It is divided into three parts:

- the analysis of variance table
- descriptive information
- information about the effect of the independent variable in the model

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Look at each of these parts separately.

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	0.00457996	0.00152665	1.96	0.1432
Error	28	0.02183054	0.00077966		
Corrected Total	31	0.02641050			

The *F* statistic and corresponding *p*-value are reported in the Analysis of Variance table. Because the reported *p*-value (0.1432) is greater than 0.05, you do not reject the null hypothesis of no difference between the means.

R-Square	Coeff Var	Root MSE	BulbWt Mean
0.173414	12.74520	0.027922	0.219082

The BulbWt Mean is the mean of all of the data values in the variable **BulbWt** without regard to **Fertilizer**.

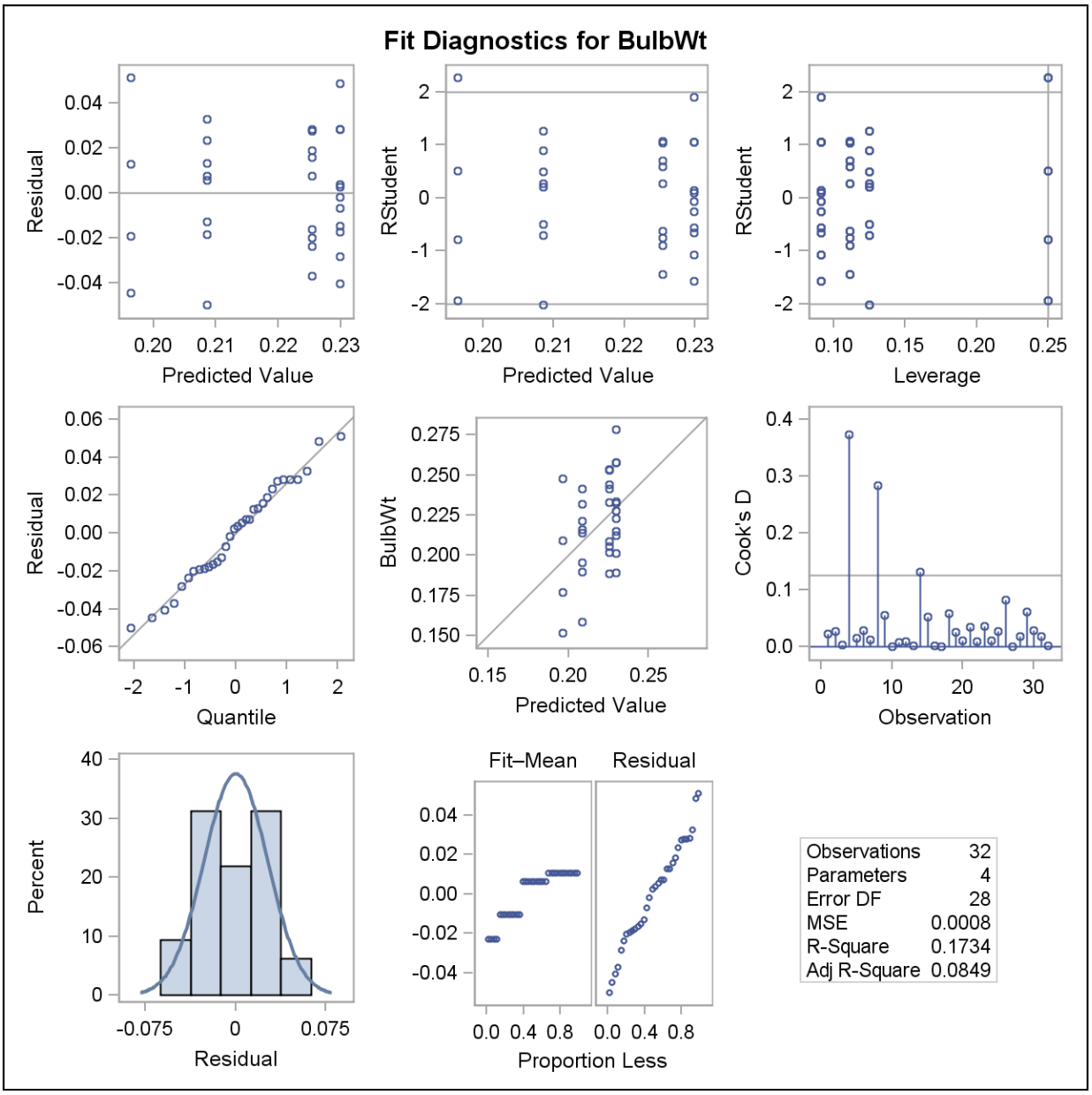
As discussed previously, the R^2 value is often interpreted as the proportion of variance accounted for by the model. Therefore, you might say that in this model, **Fertilizer** explains about 17% of the variability of **BulbWt**.

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Fertilizer	3	0.00457996	0.00152665	1.96	0.1432

For a one-way analysis of variance (only one classification variable), the information about the independent variable in the model is an exact duplicate of the model line of the analysis of variance table.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

It is good practice to check the validity of your ANOVA assumptions with the Fit Diagnostics plot.



The panel in the upper left corner shows a plot of the residuals versus the fitted values from the ANOVA model. Essentially, you are looking for a random scatter within each group. Any patterns or trends in this plot can indicate model assumption violations.

To check the normality assumption, look at the Residual Histogram and Q-Q Plot, which are at the bottom left and middle left, respectively.

The histogram has no unique peak and it has short tails. However, it is approximately symmetric.

The data values in the quantile-quantile plot stay close to the diagonal reference line and give strong support to the assumption of normally distributed errors.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

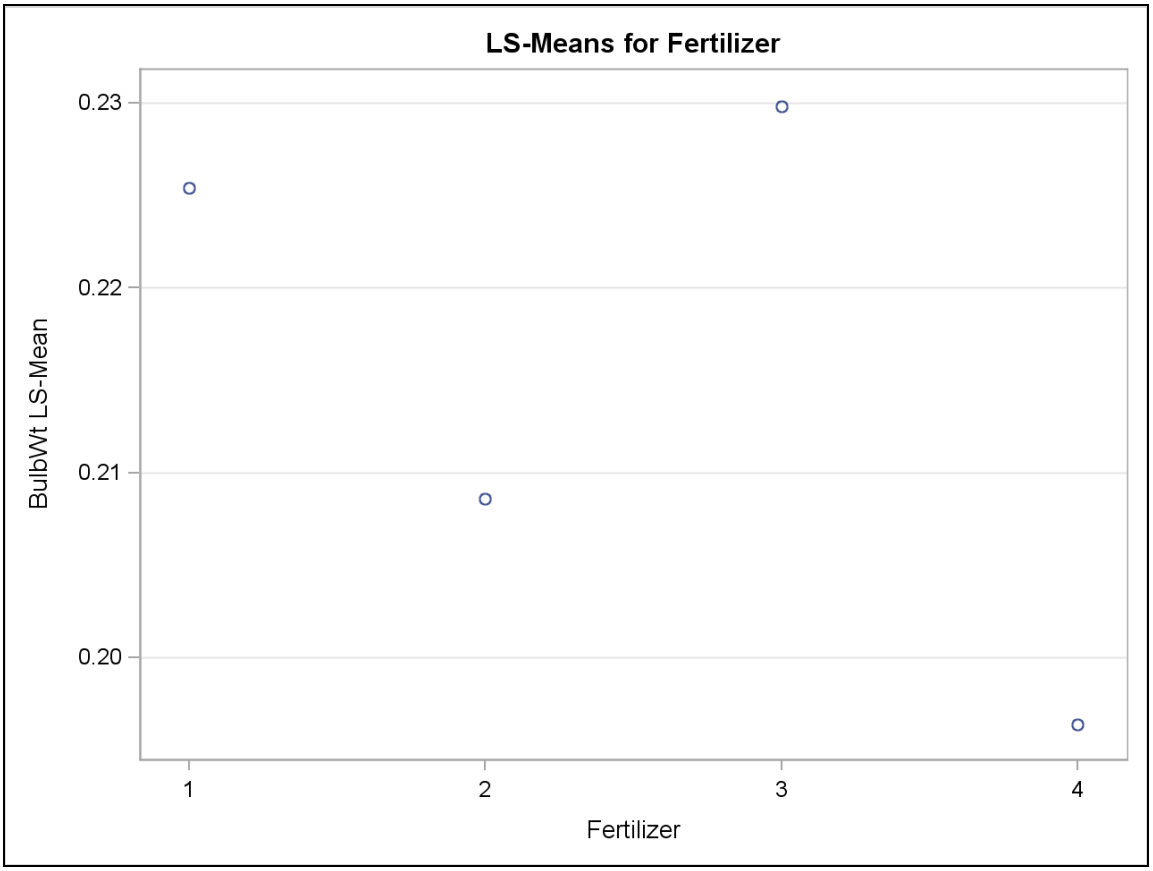
The next tables show the actual mean value comparisons among the fertilizer groups.

Fertilizer	BulbWt LSMEAN	LSMEAN Number
1	0.22540667	1
2	0.20856500	2
3	0.22982091	3
4	0.19635250	4

The fertilizer with the highest mean bulb weight is 3 and the one with the lowest is 4.

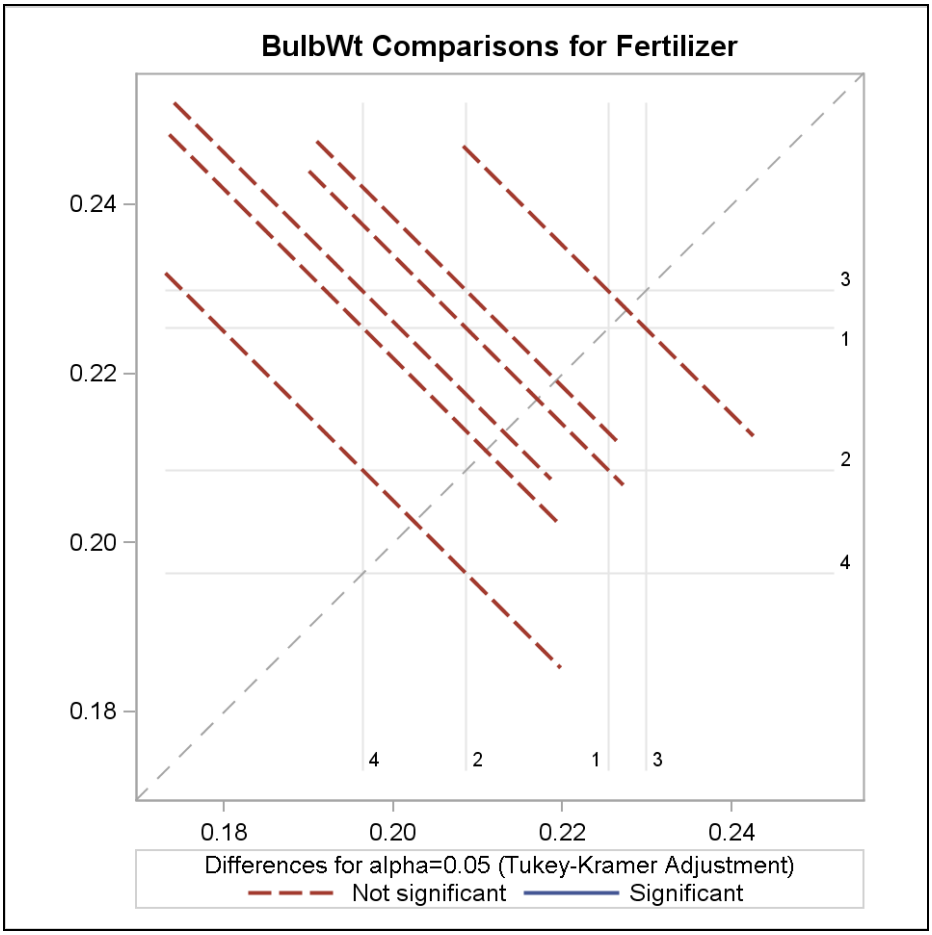
Least Squares Means for effect Fertilizer Pr > |t| for H0: LSMean(i)=LSMean(j) Dependent Variable: BulbWt				
i/j	**1**	**2**	**3**	**4**
1		0.6066	0.9848	0.3270
2	0.6066		0.3744	0.8906
3	0.9848	0.3744		0.1934
4	0.3270	0.8906	0.1934	

The *p*-values for comparing each of the group means is displayed above. The matrix displays all pairwise comparison *p*-values twice (because of the symmetry of the comparison of A*B and B*A). These *p*-values are typically consulted only when the overall test of the variable is statistically significant. Here, you have already seen that the overall test of **BulbWt** is not statistically significant.



The LS-Means plot displays the means on a dot plot.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.



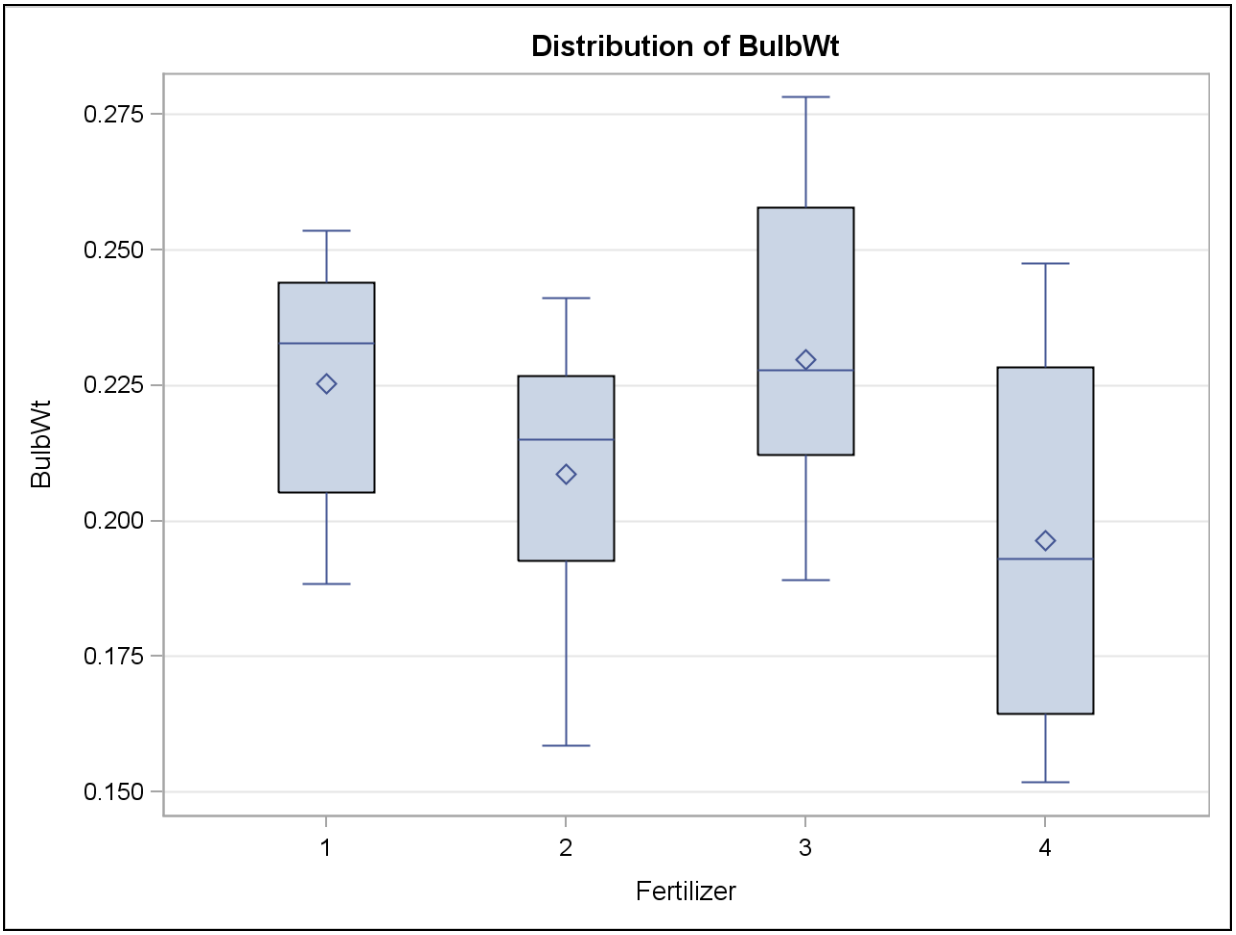
The Diffogram above displays the results of the statistical comparisons graphically. This graph is described in more detail in a subsequent example.

Near the end of the tabular output, you can check the assumption of equal variances.

Levene's Test for Homogeneity of BulbWt Variance ANOVA of Squared Deviations from Group Means					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Fertilizer	3	1.716E-6	5.719E-7	0.98	0.4173
Error	28	0.000016	5.849E-7		

The null hypothesis for the Levene's test is that the variances are equal over all **Fertilizer** groups. The *p*-value of 0.4173 is not smaller than your alpha level of 0.05 and therefore you do not reject the null hypothesis. One of your assumptions is met.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.



A set of box plots is displayed, similar to the one that you created in the previous demonstration.

Level of Fertilizer	N	BulbWt	
		Mean	Std Dev
1	9	0.22540667	0.02452242
2	8	0.20856500	0.02641979
3	11	0.22982091	0.02644365
4	4	0.19635250	0.04139664

The means are reported once more.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

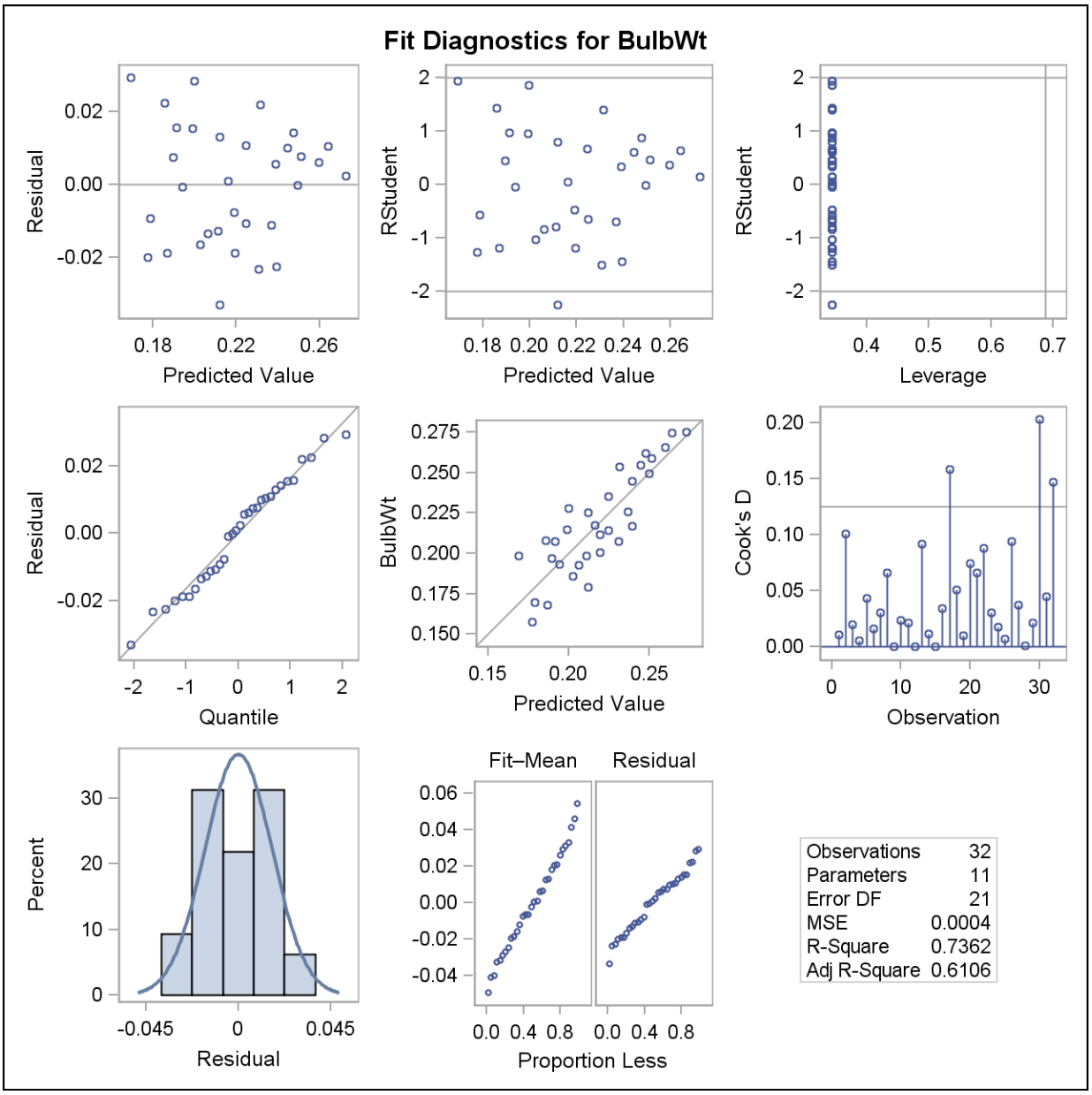
ANOVA with Blocking

Perform an ANOVA on the **MGGARLIC_BLOCK** data set to look for differences in garlic bulb weight across fertilizer types, controlling for **SECTOR**, which is the blocking variable in the design. Check for normality of the residuals.

1. Open the data table **MGGARLIC_BLOCK**, following the pattern used in previous demonstrations.
2. Select **Tasks** ⇨ **ANOVA** ⇨ **Linear Models**.
3. Under **Data**, select **BulbWeight** as the dependent variable task role and **Fertilizer** and **Sector** as the classification variables.
4. Under **Model**, select **Fertilizer** and **Sector** and then click **Main**.
5. Under **Model Options**, uncheck **Type I** and **Show parameter estimates**.
6. Click **Run**.
7. Save the EGBS project.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

A check of the normality assumption using the Q-Q Plot follows.



No severe departure from normality of the error term seems to exist.

Validation of the equal variances assumption for models with more than one independent variable is beyond the scope of this text.

Class Level Information		
Class	**Levels**	**Values**
Fertilizer	4	1 2 3 4
Sector	8	1 2 3 4 5 6 7 8

The Class Level Information table reflects the addition of the eight-level **Sector** variable.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Number of Observations Read	32
Number of Observations Used	32

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	10	0.02307263	0.00230726	5.86	0.0003
Error	21	0.00826745	0.00039369		
Corrected Total	31	0.03134008			

R-Square	Coeff Var	Root MSE	BulbWt Mean
0.736202	9.085064	0.019842	0.218398

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Fertilizer	3	0.00508630	0.00169543	4.31	0.0162
Sector	7	0.01798632	0.00256947	6.53	0.0004

The overall F-test ($F(10,21)=5.86$, $p=0.0003$) indicates that there are significant differences between the means of the garlic bulb weights across fertilizers or blocks (sectors). However, because there is more than one term in the model, you cannot tell whether the differences are due to differences among the fertilizers or differences across sectors. In order to make that determination, you must look at the subsequent tests for each factor.

What have you gained by including **Sector** in the model? If you compare the estimate of the experimental error variance (MSE), you note this is smaller compared to the data and model that included **Fertilizer** only (0.00039369 versus 0.00077966). Depending on the magnitude of the difference, this could affect the comparisons between the treatment means by finding more significant differences than the **Fertilizer**-only model, given the same sample sizes.

Also notice that the R square for this model is much greater than that in the previous model (0.736 versus 0.173). To some degree, this is a function of having more model degrees of freedom, but it is unlikely this is the only reason for this magnitude of difference.

Most important to the Montana Gourmet Garlic farmers is that the effect of **Fertilizer** in this model is now significant ($F=4.31$, $p=0.0162$). The Type III SS test is at the bottom of the output tests for differences due to each variable, controlling for (or "adjusting for") the other variable.

The Type I SS test is *sequential*. In other words, the test for each variable only adjusts for the variables above it. In this case, because the design is completely balanced, the Type I and Type III tests would be exactly the same. In general that would not be true.

In determining the usefulness of having a blocking variable (**Sector**) included in the model, you can consider the F value for the blocking variable. Some statisticians suggest that if this ratio is greater than 1, then the blocking factor is useful. If the ratio is less than 1, then adding the variable is detrimental to the analysis. If you find that including the blocking factor is detrimental to the analysis, then you can exclude it from future studies, but it ***must*** be included in all ANOVA models calculated with the sample that you already collected. This is because blocking places a restriction on the random assignment of units to treatments, and modeling the data without the blocking variable treats the data as if that restriction did not exist.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Post Hoc Pairwise Comparisons

Request post-hoc comparison tests to look for pairwise differences among **Fertilizer** groups.

1. Re-open the previous Linear Models task by right-clicking the icon for it and selecting **Modify** from the drop-down menu.

2. Select **Least Squares** under Post Hoc Tests and then click **Add** under the white space for **Effects to estimate**.

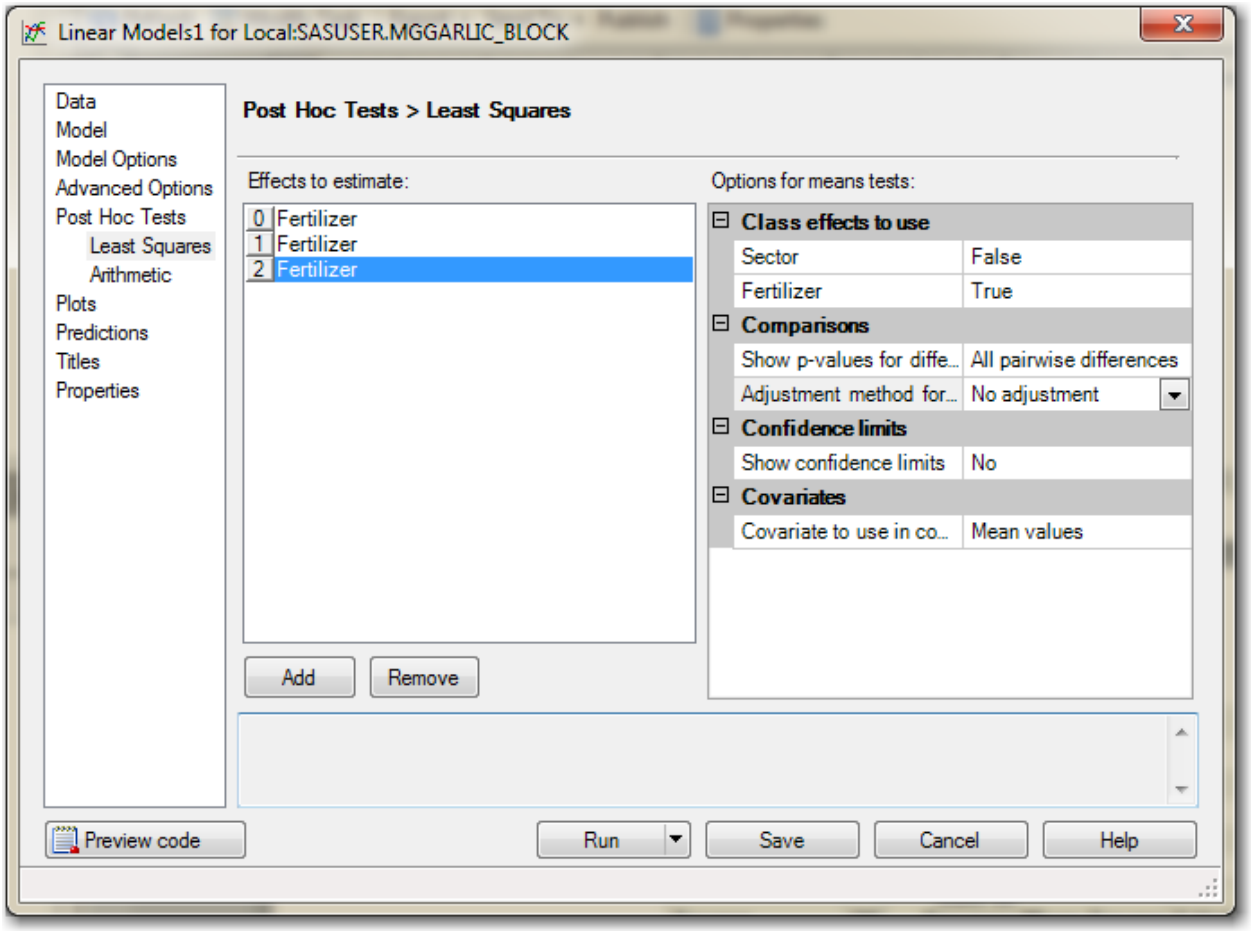
3. Click **False** next to **Fertilizer** from the **Options for means tests** and then click that appears next to it. Change this value to **True** by scrolling to that value and clicking it.

4. In a similar fashion, under **Comparisons**, change the **Show p-values for differences** option from **None** to **All pairwise differences**. The **Adjustment method for comparison** can be selected as **Tukey** or left as **Default** because Tukey is the default method used when **All pairwise differences** is chosen above.

5. Click **Add** under the white space for **Effects to estimate** and effect 1 will appear. Select **Fertilizer** again as the **Class effects to use** variable. This time the choice for **Show p-values for differences** should be **Control using first level**. The default **Adjustment method** will be **Dunnett** and the default control group will be **1**. You will want to change that at a later step.

6. Click **Add** under the white space for **Effects to estimate** and effect 2 will appear. Select **Fertilizer** again as the **Class effects to use** variable. Change the **Show p-values for differences** option from **None** to **All pairwise differences**. The **Adjustment method for comparison** can be selected as **No adjustment** this time. This means that the tests will not control for experimentwise error rate (in other words, there will be no multiple tests adjustment).

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

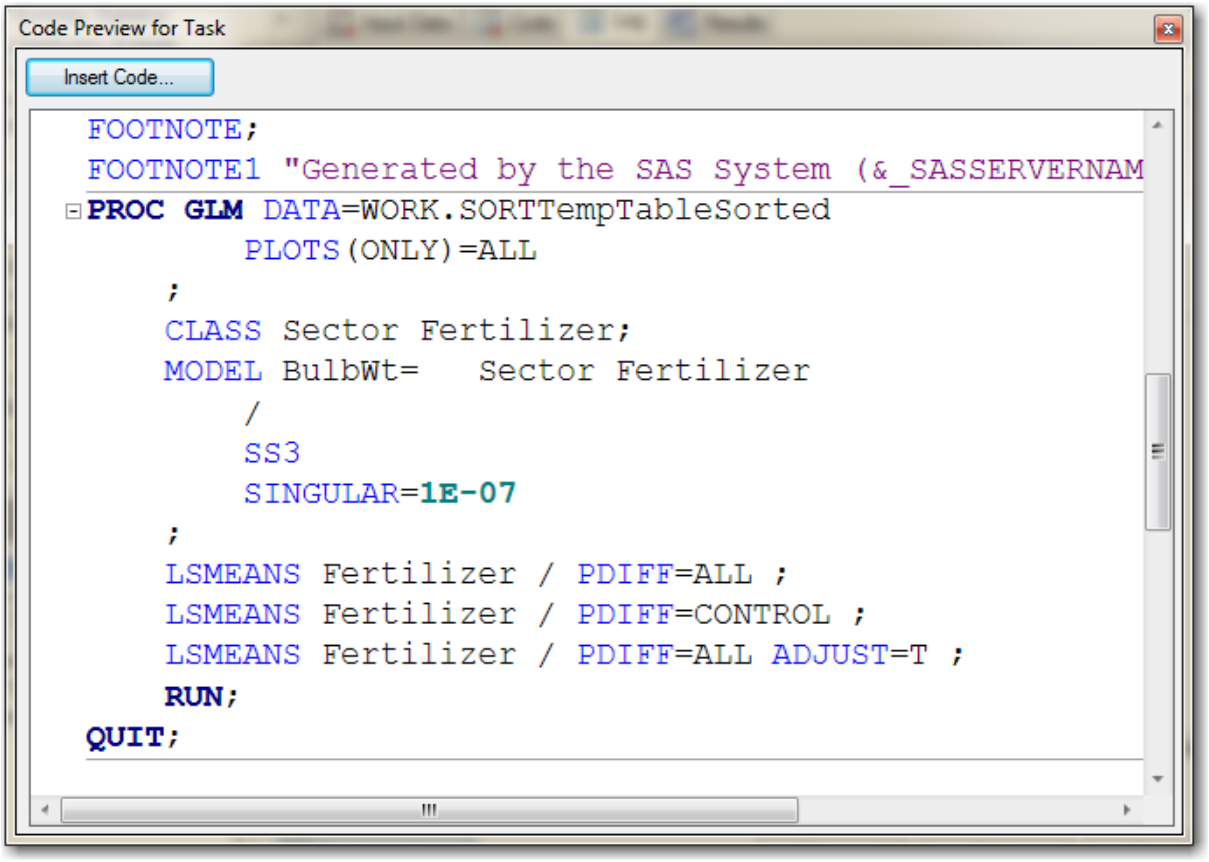


In order to select a different group for the control group for the Dunnett comparisons, the code needs to be adjusted slightly.

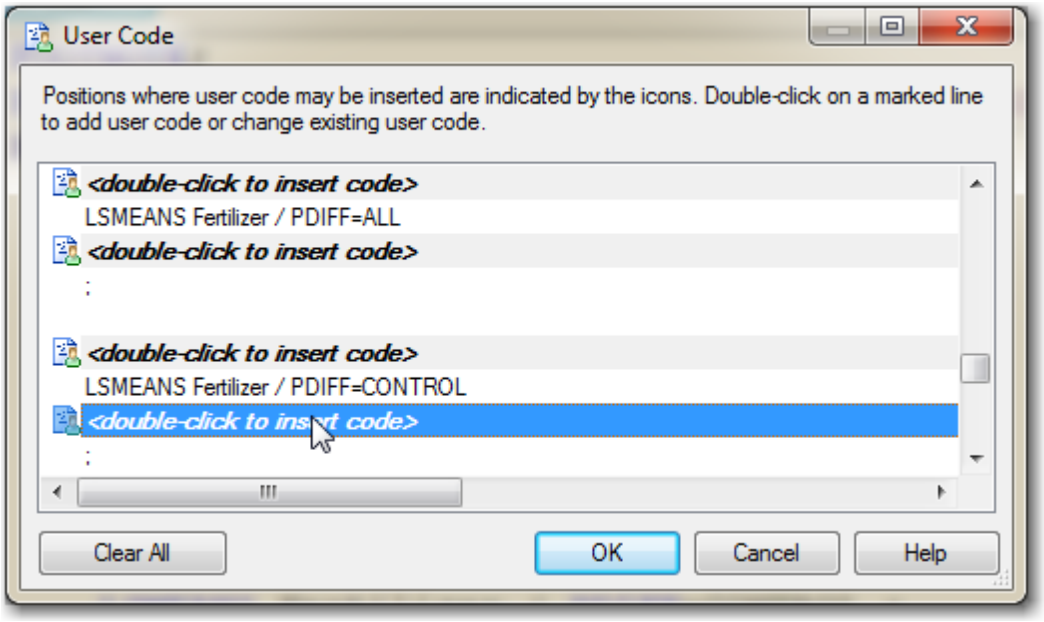
7. Click **Preview code**.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

8. Click **Insert Code** in the pop-up window.

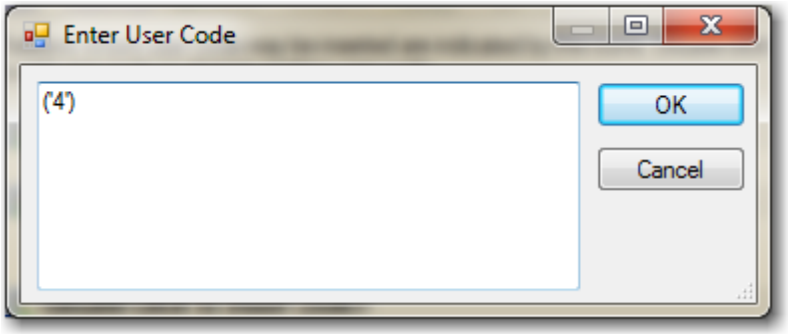


9. Scroll down the place in the code just under the words **LSMEANS Fertilizer / PDIFF=CONTROL** and double-click to insert code, as instructed.



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

10. Type **('4')** in the window that appears. Level 4 is the chemical fertilizer and is used as the control group. The others are organic. By default, the first level of the classification variable is used as the control.



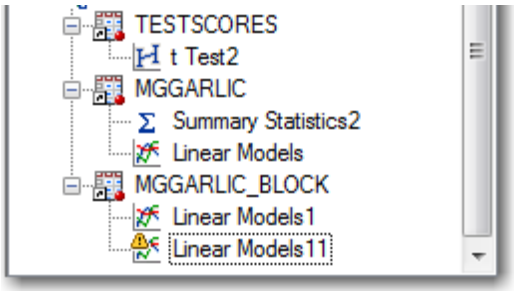
11. Click **OK** twice to close the last two windows that were opened. Notice that if you scroll down to the place in the Code Preview for Task window corresponding to where you just inserted code, then you see your additional code there.

```
LSMEANS Fertilizer / PDIFF=ALL ;
LSMEANS Fertilizer / PDIFF=CONTROL
/* Start of custom user code. */
('4')
/* End of custom user code. */
;
```

12. Click in the upper right corner to close that window.

13. Click **Run**. Do not replace the previous results.

    Notice that the task icon in the Project Tree pane has a yellow triangle.



The log shows the reason for the yellow triangle. There is a warning:

```
57          LSMEANS Fertilizer / PDIFF=ALL ADJUST=T ;
58          RUN;

WARNING: ADJUST=T implies no adjustment for simultaneous inference.
```

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

When all pairwise comparisons are tested with no adjustment for multiple comparisons, SAS issues a warning just to make sure that this is what you want. Typically you will adjust for multiple comparisons, but the unadjusted tests are shown for comparison.

The following is selected output is for the Tukey comparisons.

Fertilizer	BulbWt LSMEAN	LSMEAN Number
1	0.23625000	1
2	0.21115125	2
3	0.22330125	3
4	0.20288875	4

Least Squares Means for effect Fertilizer
Pr > |t| for H0: LSMean(i)=LSMean(j)

Dependent Variable: BulbWt

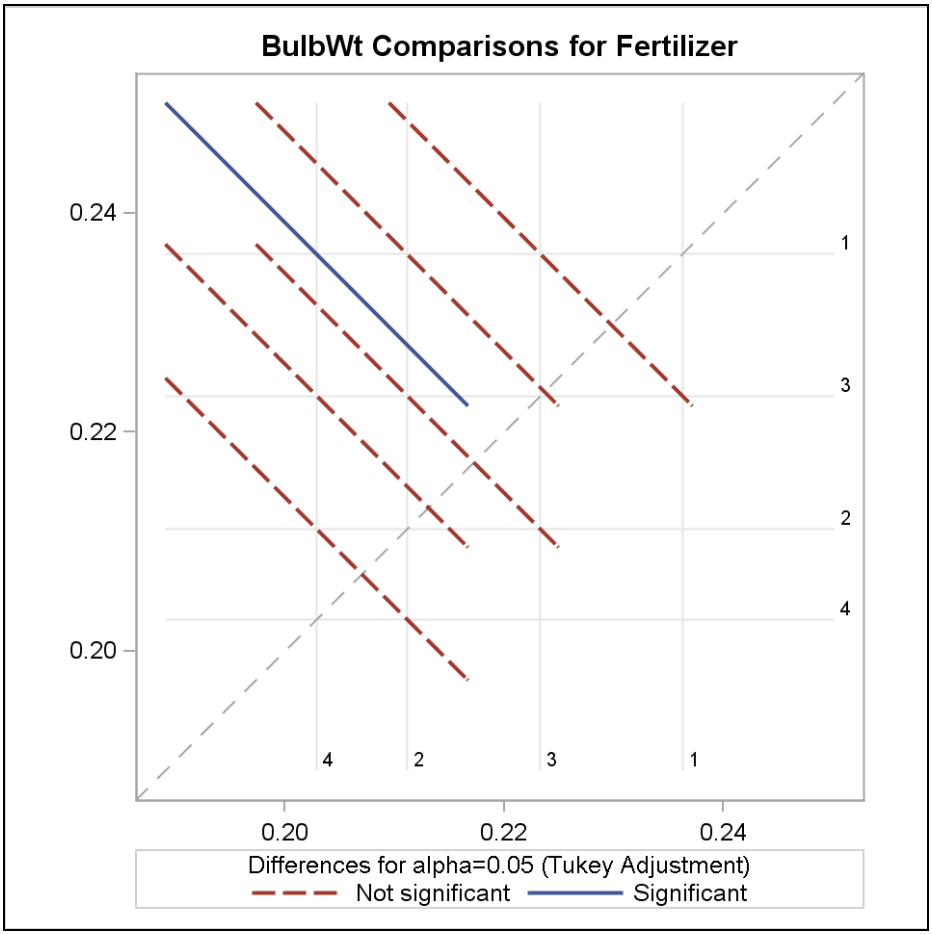
i/j	1	2	3	4
1		0.0840	0.5699	0.0144
2	0.0840		0.6186	0.8383
3	0.5699	0.6186		0.1995
4	0.0144	0.8383	0.1995	

The first part of the output shows the means for each group. The second part of the output shows *p*-values from pairwise comparisons of all possible combinations of means. Notice that row 2/column 4 has the same *p*-value as row 4/column 2 because the same two means are compared in each case. Both are displayed as a convenience to the user. Notice also that row 1/column 1, row 2/column 2, and so on, are blank, because it does not make any sense to compare a mean to itself.

The only significant pairwise difference is between fertilizer 1 and fertilizer 4 (*p*-value=0.0144).

The Least Square Means are shown graphically in the mean plot. The Tukey-adjusted differences among the least squared means are shown in the diffogram.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.



The solid line shows the significant difference between fertilizers 1 and 4. (The confidence limit for the difference does not cross the diagonal equivalence line.)

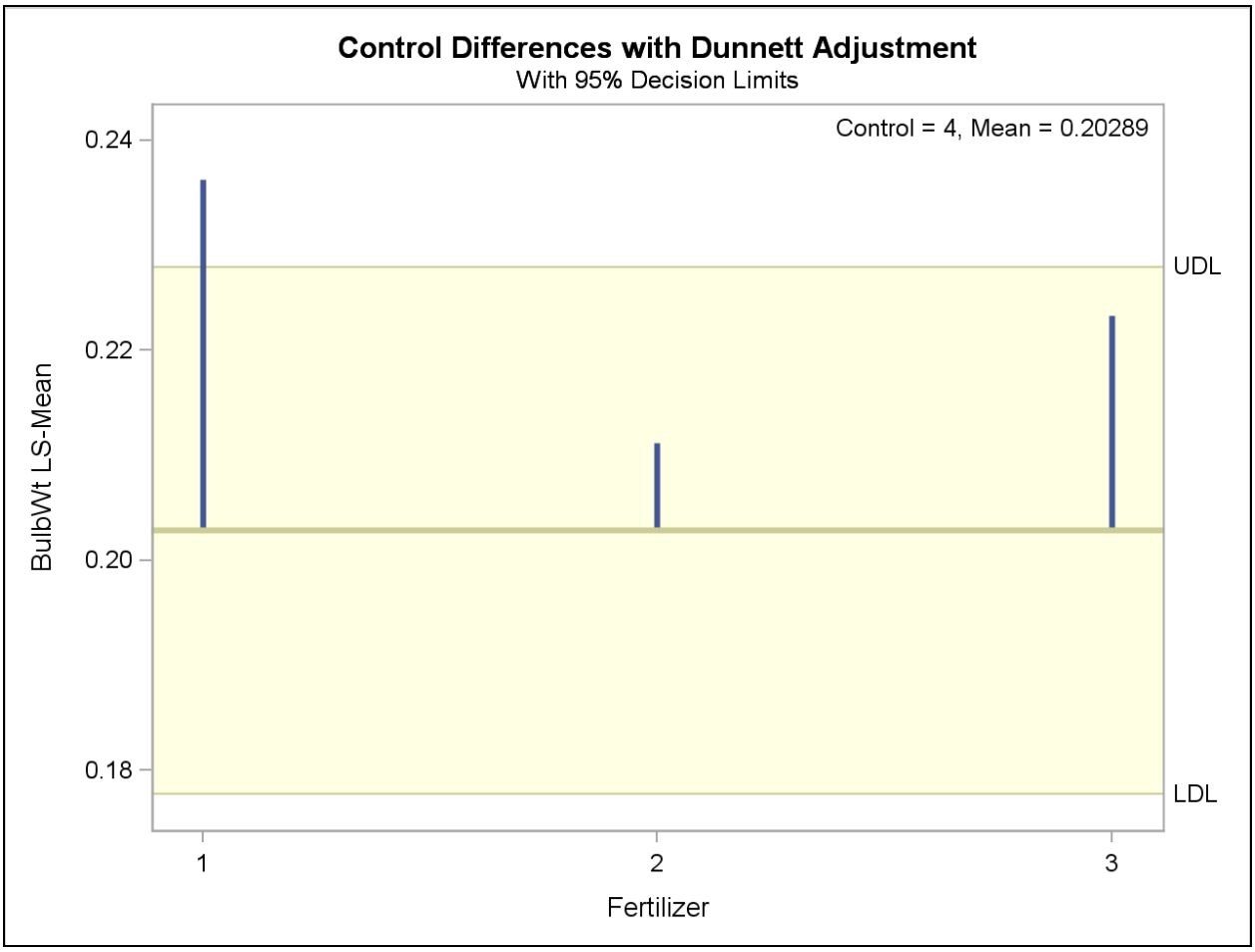
The following output is for the Dunnett comparisons against a control:

Fertilizer	BulbWt LSMEAN	H0:LSMean=Control Pr > |t|
1	0.23625000	0.0080
2	0.21115125	0.7435
3	0.22330125	0.1274
4	0.20288875	

In this case, the first three fertilizers are compared to fertilizer 4, the chemical fertilizer. Even though the mean weights of garlic bulbs using any of the three organic methods are all greater than the mean weight of garlic bulbs grown using the chemical fertilizer, only fertilizer 1 can be said to be statistically significantly different from the control.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

The Control Plot is below:



This plot corresponds to the tables that were summarized. The horizontal line is drawn at the least squared mean for group 4, which is 0.20289. The three other means are represented by the tops of the vertical lines extending from the horizontal control line. The only line that extended beyond the shaded area of nonsignificance is the line for fertilizer 1. That shows graphically that the mean bulb weight for fertilizer 1 is significantly different from the mean bulb weight for fertilizer 4.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Finally, for comparison, the *t*-tests are shown, which do not adjust for multiple comparisons and are therefore more liberal than tests that do control for experimentwise error:

Fertilizer	BulbWt LSMEAN	LSMEAN Number
1	0.23625000	1
2	0.21115125	2
3	0.22330125	3
4	0.20288875	4

Least Squares Means for effect Fertilizer
Pr > |t| for H0: LSMean(i)=LSMean(j)

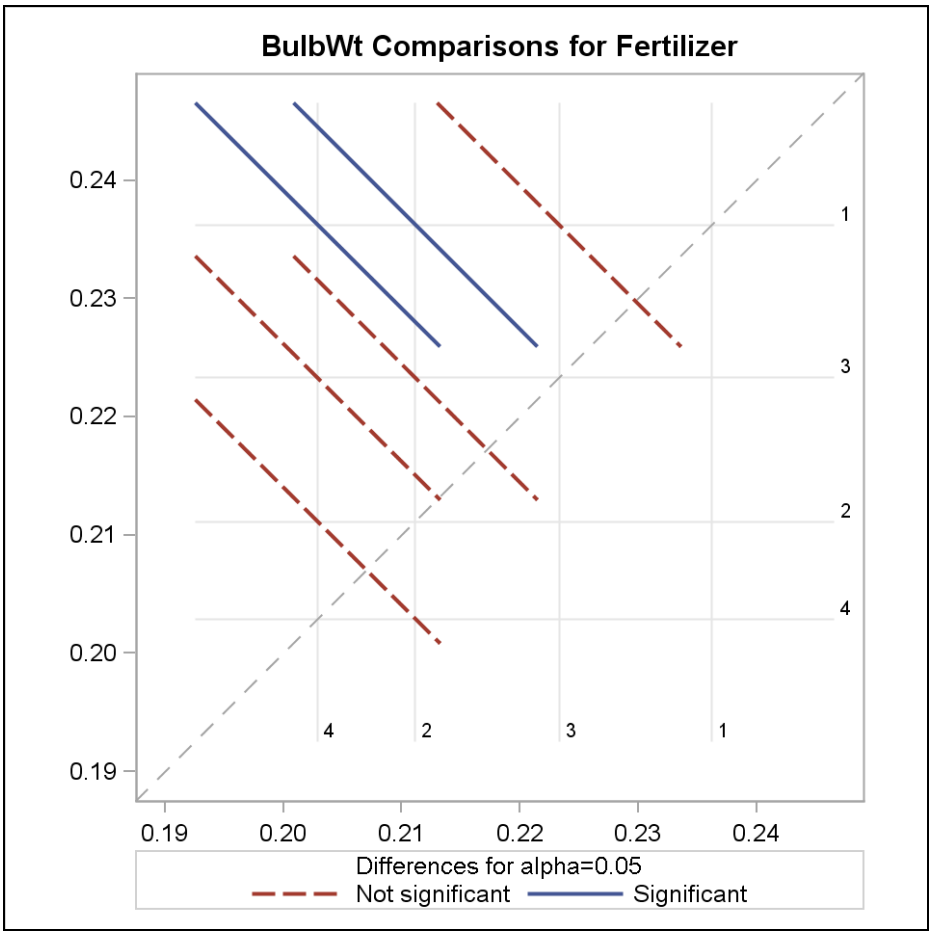
Dependent Variable: BulbWt

i/j	1	2	3	4
1		0.0195	0.2059	0.0029
2	0.0195		0.2342	0.4143
3	0.2059	0.2342		0.0523
4	0.0029	0.4143	0.0523	

The *p*-values in this table are all smaller than those in the Tukey table. In fact, using this method shows one additional significant pairwise difference. Fertilizer 1 is significantly different from fertilizer 2 (p=0.0195). The comparison between 3 and 4 is nearly significant at alpha=0.05 (p=0.0523).

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

The diffogram shows the additional significant difference:



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Two-Way ANOVA with Interactions

Before conducting an analysis of variance, you should explore the data.

Presume that the initial data exploration was completed (output not shown here) and that no particular concerns were noted about unusual data values or the distribution of the data. During this exploration, you determine that the sample sizes for all treatments are not equal. The researchers recruited 240 patients (80 per heart disease category), but only 170 were randomized into the trial.

1. Open the **DRUG** data set.

DRUG

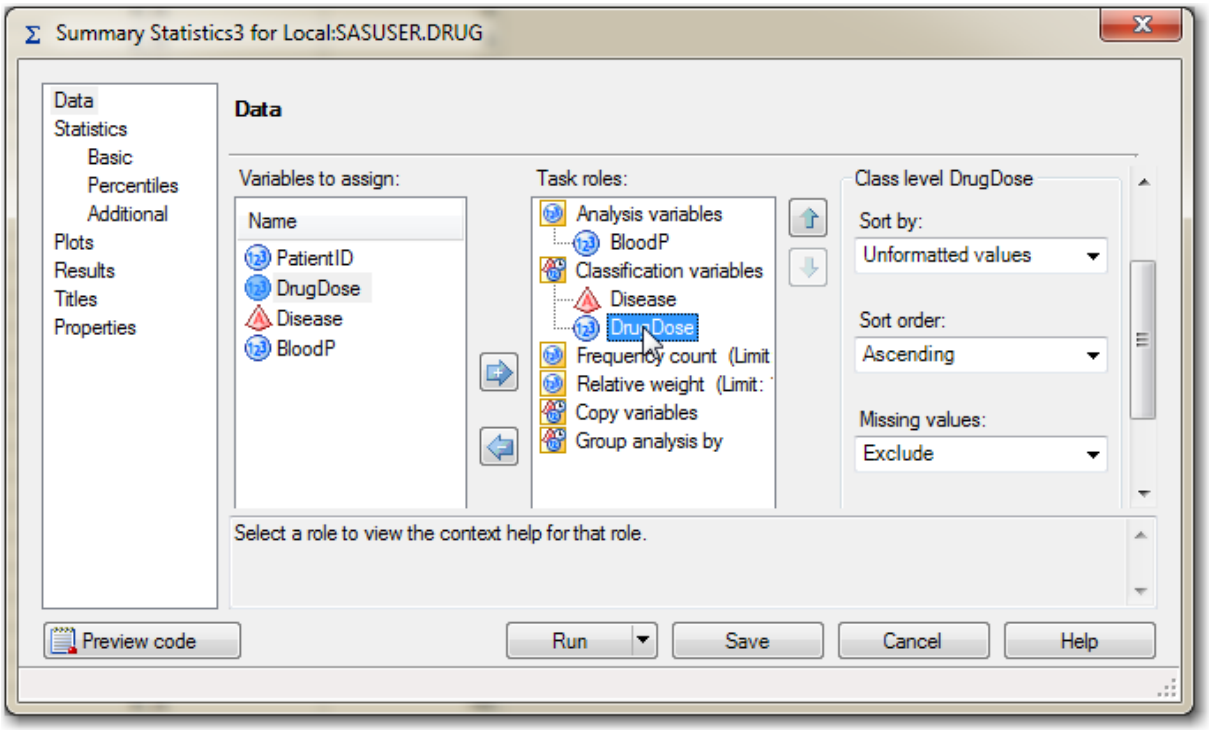
Filter and Sort | Query Builder | Data | Describe | Graph | Analyze | Export | Send To

	PatientID	DrugDose	Disease	BloodP
1	69	2	B	13
2	162	4	A	-47
3	181	1	B	12
4	209	4	A	-4
5	308	2	A	4
6	331	4	C	37
7	340	4	C	-19
8	350	1	B	-9
9	360	2	B	-17
10	363	4	A	-41

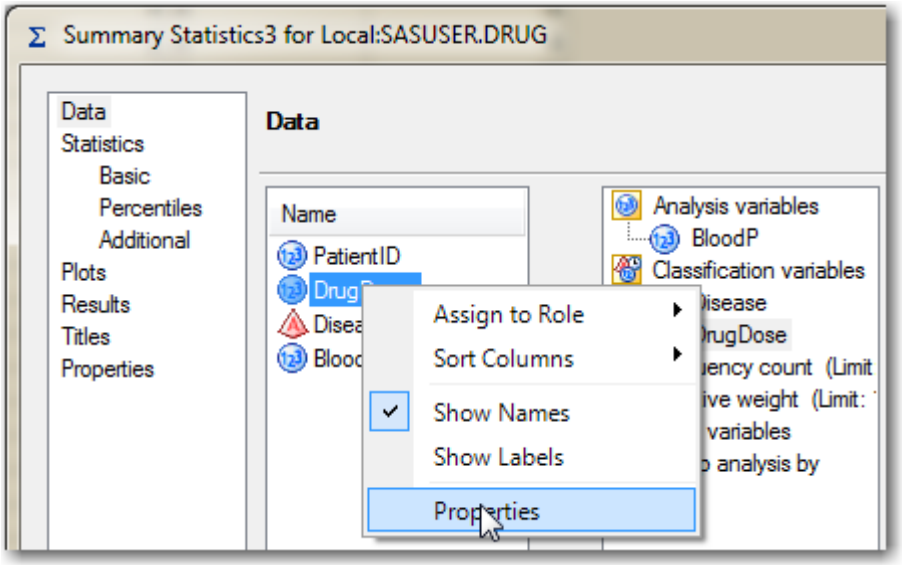
Negative values for **BloodP** mean that diastolic blood pressure was reduced by that amount. Positive values mean that the **BloodP** increased.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

2. Use the Summary Statistics task, choosing **BloodP** for the analysis variables role and **Disease** and **DrugDose** for the classification variables role.

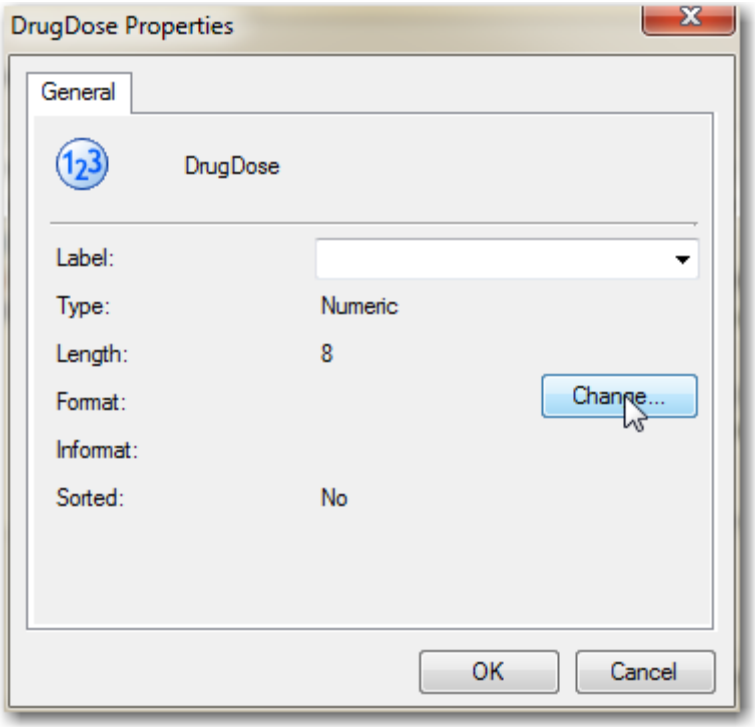


3. Assign a user define format to the **DrugDose** variable by right-clicking **DrugDose** in the **Variables to assign** pane and selecting **Properties**.

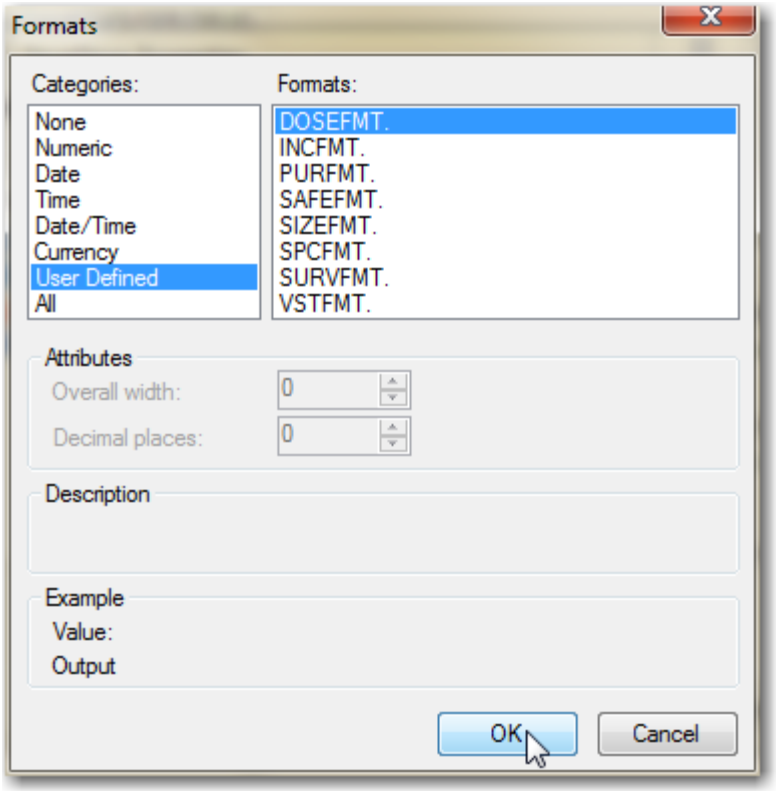


Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

4. In the Properties window, click **Change** next to **Format**.



5. Select **User Defined** at the left and **DOSEFMT** at the right and then click **OK** twice to close the last two windows.



Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

6. Run the task.

Analysis Variable : BloodP							
Disease	**DrugDose**	**N Obs**	**Mean**	**Std Dev**	**Minimum**	**Maximum**	**N**
A	Placebo	12	1.3333333	13.5333483	-22.0000000	25.0000000	12
	50 mg	16	-9.6875000	18.8881577	-37.0000000	19.0000000	16
	100 mg	13	-26.2307692	18.1390640	-51.0000000	11.0000000	13
	200 mg	18	-22.5555556	21.0970369	-61.0000000	12.0000000	18
B	Placebo	15	-8.1333333	16.9109714	-39.0000000	22.0000000	15
	50 mg	15	5.4000000	21.8886794	-45.0000000	35.0000000	15
	100 mg	14	24.7857143	23.7427838	-24.0000000	60.0000000	14
	200 mg	13	23.2307692	23.5872630	-22.0000000	55.0000000	13
C	Placebo	14	0.4285714	20.2929100	-38.0000000	45.0000000	14
	50 mg	13	-4.8461538	24.0341637	-36.0000000	50.0000000	13
	100 mg	14	-5.1428571	13.9827209	-26.0000000	27.0000000	14
	200 mg	13	1.3076923	28.7847894	-57.0000000	42.0000000	13

The means do seem to vary from group to group. However, it is not obvious if there is a consistent trend across different levels of **DrugDose**. It seems that the trend of **BloodP** across levels of **DrugDose** might be different from disease to disease.

You can use the Linear Models task to discover whether these differences and their interactions are statistically significant.

To test the hypothesis that the effect of **DrugDose** differs across diseases, perform a 2-way ANOVA with an interaction.

1. Select **Tasks** ⇨ **ANOVA** ⇨ **Linear Models**.

2. Under **Data**, select **Bloodp** as the dependent variable task role and **Disease** and **DrugDose** as the classification variables role.

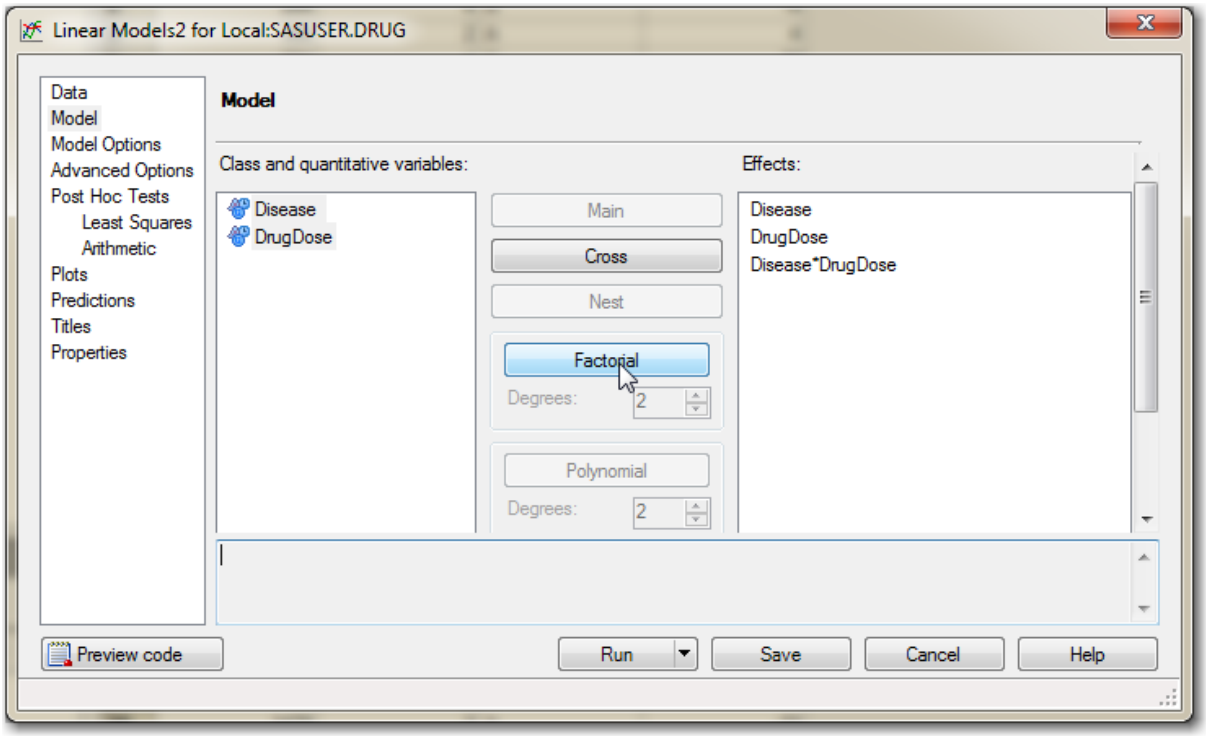
   DrugDose is saved in SAS as numeric variable and could have been assigned to the quantitative variables role. However, the scale of measurement of the variable is not continuous, but rather ordinal, so it should be added as a classification variable.

3. Assign the **DOSEFMT** format to the **DrugDose** variable, as in the previous task.

4. Select **Model** at the left and highlight both **Disease** and **DrugDose** in the **Class and quantitative variables** pane (by clicking one and then holding down the CTRL key on the keyboard and then clicking the other variable).

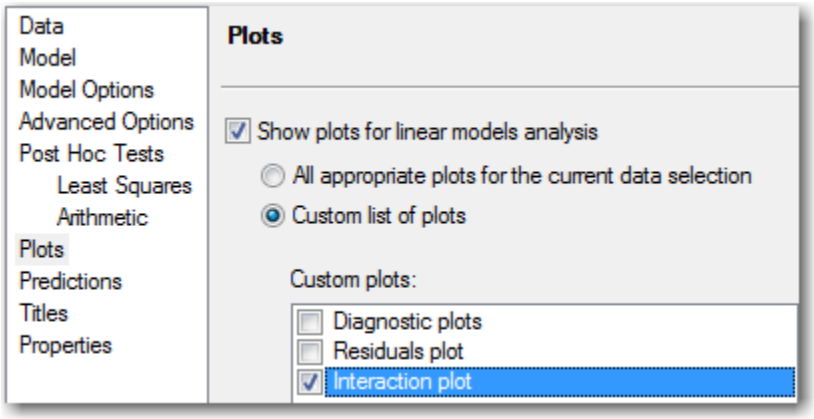
Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

5. Click **Factorial** and see the terms, **DrugDose**, **Disease**, and **DrugDose*Disease** displayed in the **Effects** pane.



6. Under **Model Options**, deselect **Show parameter estimates**.

7. Under **Plots**, click the radio button for **Custom list of plots**, deselect **Diagnostic plots,** and select **Interaction plot**.

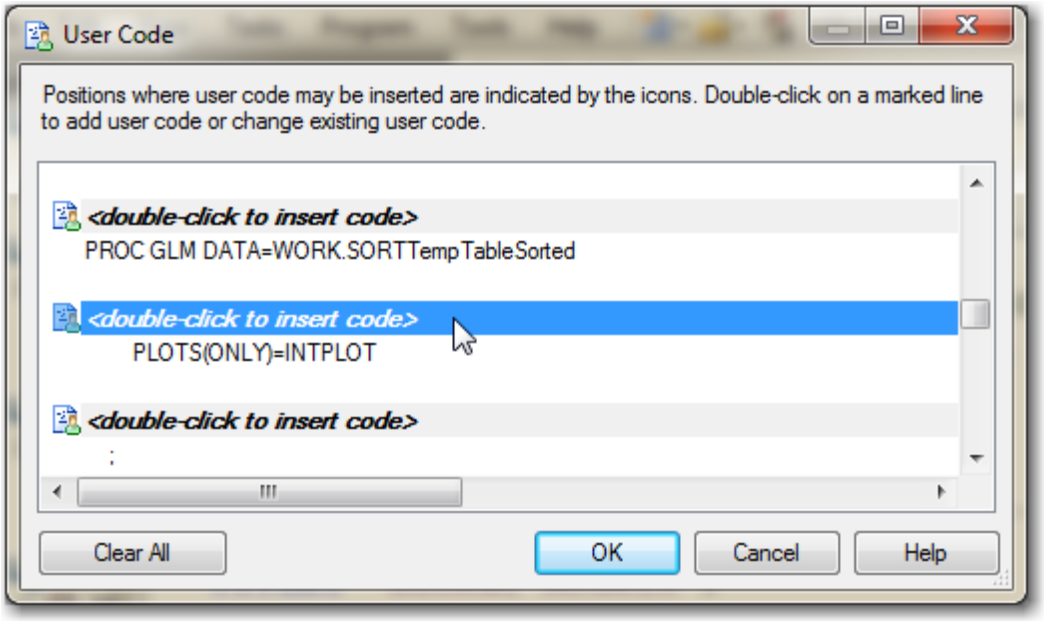


8. Open the programming code by clicking **Preview code**.

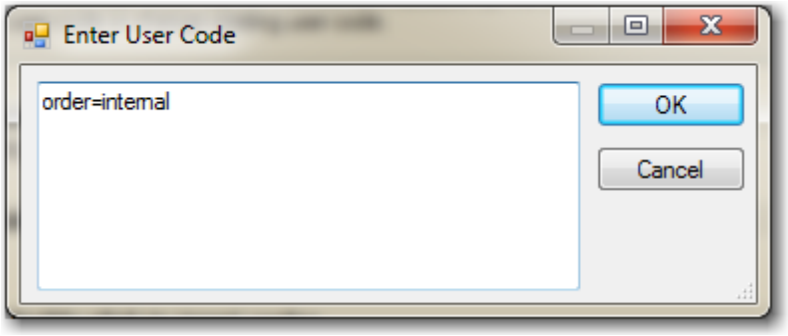
9. Click **Insert Code**.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

10. Locate the code just below the PROC GLM statement and double-click the note ***<double-click to insert code>***.



11. In the text box, type the following:



12. Click **OK** twice to close the last two windows and then click to close the Code Preview for Task window.

13. Click **Run**.

    Linear Models task output:

Class Level Information		
Class	**Levels**	**Values**
DrugDose	4	Placebo 50 mg 100 mg 200 mg
Disease	3	A B C

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Number of Observations Read	170
Number of Observations Used	170

The next part of the output shows the source table with the *F*-test for the overall model. This tests the null hypothesis that none of the effects in the model is statistically different. In other words, that there are no differences among the 12 group means (one for each **DrugDose** * **Disease** combination).

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	36476.8353	3316.0759	7.66	<.0001
Error	158	68366.4589	432.6991		
Corrected Total	169	104843.2941			

R-Square	Coeff Var	Root MSE	BloodP Mean
0.347918	-906.7286	20.80142	-2.294118

The BloodP Mean value indicates that the average blood pressure change over all observations is -2.294118. (the same value as would be obtained using the Summary Statistics task.) The R-Square for this model is 0.347918.

The *p*-value is <0.0001. Presuming an alpha equal to 0.05, you reject the null hypothesis and conclude that at least one treatment mean is different from one other treatment mean. Which factor(s) explain this difference?

The next part of the output shows tests of the main effects and the interaction.

Source	DF	Type I SS	Mean Square	F Value	Pr > F
DrugDose	3	54.03137	18.01046	0.04	0.9886
Disease	2	19276.48690	9638.24345	22.27	<.0001
DrugDose*Disease	6	17146.31698	2857.71950	6.60	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
DrugDose	3	335.73526	111.91175	0.26	0.8551
Disease	2	18742.62386	9371.31193	21.66	<.0001
DrugDose*Disease	6	17146.31698	2857.71950	6.60	<.0001

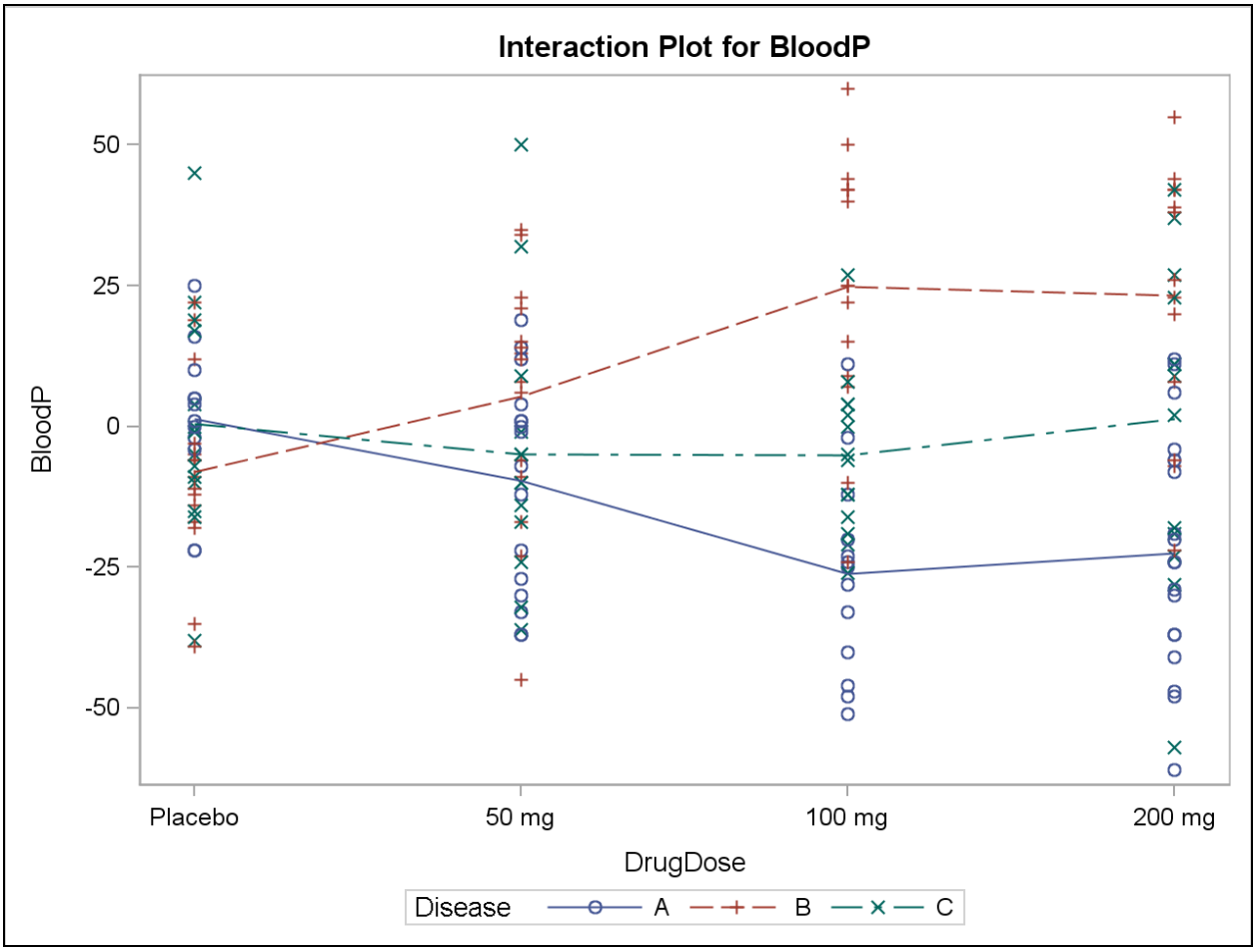
The Type I SS are *model-order dependent*. Each effect is adjusted only for the preceding effects in the model. They are also known as *sequential sums of squares*. They are useful in cases where the marginal (additional) effect for adding terms in a specific order is important. An example is a test of polynomials, where X, X*X, and X*X*X are in the model. Each term is tested only controlling for a lower-order term. The TYPE I SS values are additive. They sum to the Model Sum of Squares for the overall model.

The Type III sums of squares are commonly called *partial sums of squares*. The Type III sum of squares for a particular variable is the increase in the model sum of squares due to adding the variable to a model that already contains all the other variables in the model. Type III sums of squares, therefore, do not depend on the order in which the explanatory variables are specified in the model. The Type III SS values are not generally additive (except in a completely balanced design). The values do not necessarily sum to the Model SS.

You generally interpret and report results based on the Type III SS.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

You should consider the test for the interaction first, because if there is an interaction, then by definition this means that the effect of each main effect is different at each level of the other main effect. The *p*-value for **DrugDose*Disease** is <0.0001. Presuming an alpha of 0.05, you reject the null hypothesis. You have sufficient evidence to conclude that there is an interaction between the two factors. As shown in the graph, the effect of the level of drug changes for different disease types.



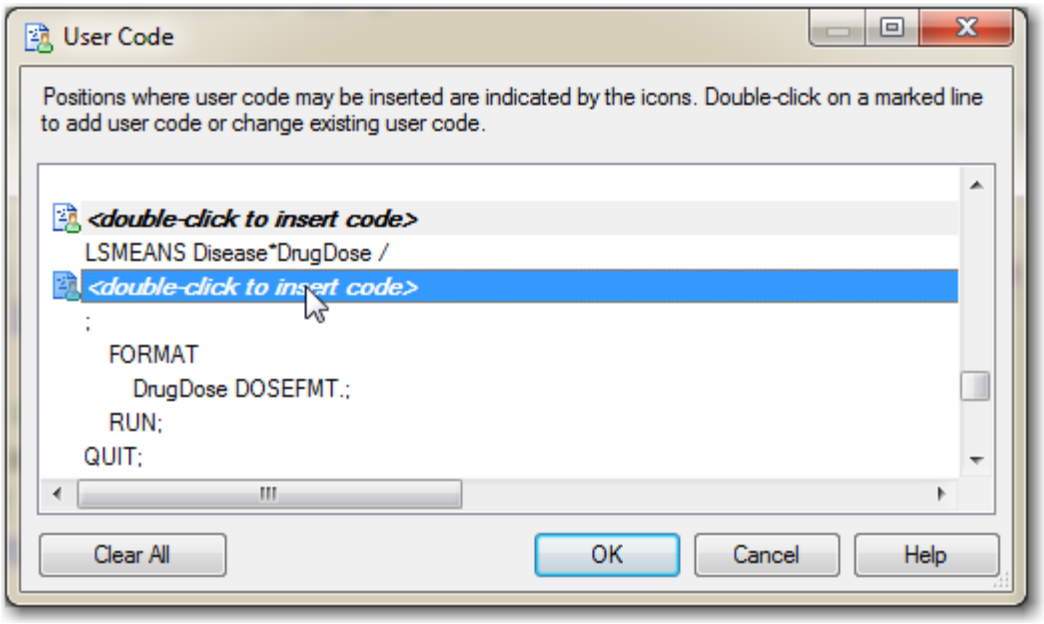
Optional:

Because of the interaction, you do not know the effect of **DrugDose** at any particular level of **Disease**. Run a post-hoc test of **DrugDose** at each level of **Disease**.

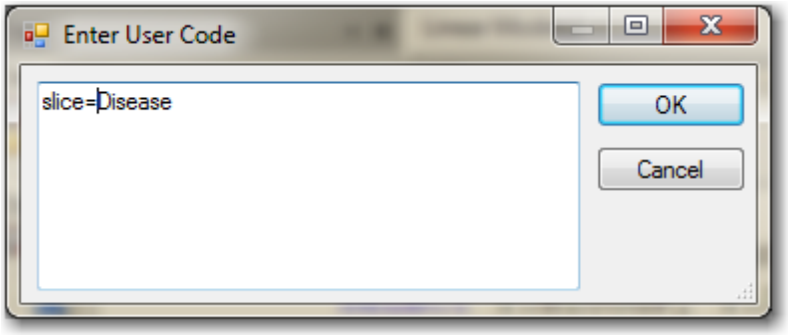
1. Reopen the previous Linear Models task window by right-clicking its title and selecting **Modify** from the drop-down menu.
2. Select **Least Squares** under **Post Hoc Tests** and then click **Add** under the white space for **Effects to estimate**.
3. Click **False** next to **Disease*DrugDose** from the **Options for means tests** and then click that appears next to it. Change this value to **True** by scrolling to that value and clicking it.
4. Open the programming code by clicking **Preview code**.
5. Click **Insert Code**.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

6. Locate the code just below the LSMEANS statement and double-click the note ***<double-click to insert code>***.



7. In the text box, type the following:



8. Click **OK** twice to close the last two windows and then click to close the Code Preview for Task window.

9. Click **Run**.

10. Replace the results from the previous run.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Selected LSMEANS statement option:

SLICE= specifies effects within which to test for differences between interaction LS-mean effects. This can produce what are known as *tests of simple effects* (Winer 1971). For example, suppose that A*B is significant and you want to test for the effect of A within each level of B. The appropriate LSMEANS statement is as follows:

```
lsmeans A*B / slice=B;
```

Selected Output

The first table is a report of the least squared mean for each unique **DrugDose** * **Disease** combination.

DrugDose	Disease	BloodP LSMEAN
Placebo	**A**	1.3333333
Placebo	**B**	-8.1333333
Placebo	**C**	0.4285714
50 mg	**A**	-9.6875000
50 mg	**B**	5.4000000
50 mg	**C**	-4.8461538
100 mg	**A**	-26.2307692
100 mg	**B**	24.7857143
100 mg	**C**	-5.1428571
200 mg	**A**	-22.5555556
200 mg	**B**	23.2307692
200 mg	**C**	1.3076923

The second table displays a test of the effect of **DrugDose** at each level of **Disease**.

DrugDose*Disease Effect Sliced by Disease for BloodP					
Disease	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
A	3	6320.126747	2106.708916	4.87	0.0029
B	3	10561	3520.222833	8.14	<.0001
C	3	468.099308	156.033103	0.36	0.7815

The **DrugDose** effect is significant when used in patients with either disease A or disease B, but not in patients with disease C.

Given all of this information, it seems that you would want to aggressively treat blood pressure in people with disease A with high doses of the drug. For those with disease B (perhaps caused by a traumatic event), treating with the drug at all would be a mistake. For those with disease C, there seems to be no effect on blood pressure.

Copyright © 2011, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.